

Review article

Reconfigurable battery System Topology, Operation and Feature: A Comprehensive Survey

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ABSTRACT

With the increasing application of energy storage systems in modern power system, the development of battery systems offering flexible control and enhanced fault tolerance has become increasingly important. Among various approaches, reconfigurable battery systems (RBS) have attracted considerable attention due to their ability to dynamically reconfigure connections at the cell level, thereby providing high flexibility and strong fault-tolerance capabilities. To date, numerous reconfigurable battery topologies have been proposed. These systems enable adaptive adjustment of voltage, capacity, and power output through dynamic series-parallel reconfiguration of battery cells or modules, as well as flexible adjustment of power conversion topologies. This paper presents a classification of existing RBS topologies based on the number of switches per cell and provides a comprehensive review of these configurations, explicitly specifying the number of switching devices required in each topology. Furthermore, a detailed comparative analysis of key performance metrics is conducted, and the suitable application scenarios for different topologies are identified. Finally, based on this analysis, potential future research directions for large-scale energy storage system architectures are discussed.

1. Introduction

In the context of global electrification, renewable energy sources have been widely adopted in power generation worldwide [1]. However, despite their significant environmental benefits, the inherent intermittency and volatility of renewable energy complicate reliable, all-weather power supply [2]. Battery energy storage systems (BESS) play a critical role in addressing this challenge by balancing supply and demand and enhancing grid stability. To fulfill these functions effectively, BESS must offer high safety, low cost, high efficiency, and long cycle life [3,4]. As the penetration of battery energy storage increases in electric vehicles (EVs), hybrid electric vehicles (HEVs), and backup power systems, the design and operation of these systems face growing challenges [5]. The expanding scale of energy storage, combined with increasingly variable operating conditions, demands battery systems with improved cell balancing capabilities, faster charge/discharge rates, and higher fault tolerance.

Nevertheless, traditional fixed-topology battery systems often suffer from cell-to-cell variations caused by manufacturing tolerances and uneven operational conditions. These inconsistencies lead to accelerated

battery degradation, imbalanced charge/discharge processes, and reduced usable capacity. As a result, the service life of the battery pack is shortened and overall system efficiency is lowered. Progressive capacity fade further limits the effective use of available energy storage, causing energy inefficiency. Moreover, the failure of one or more cells can seriously degrade performance and, in severe cases, result in complete system failure. With the growing demand for advanced battery systems and the rapid progress in energy storage technology, traditional fixed battery designs are increasingly unable to meet the efficiency and adaptability requirements of future energy infrastructures. To overcome these limitations, reconfigurable battery systems (RBS) have received considerable attention in recent years. The integration of switching devices into the battery management system (BMS) enables RBS to reach higher efficiency and longer cycle life. It also lowers the environmental impact across the entire lifecycle. All these benefits are critical for sustainable large-scale energy storage. Furthermore, RBS technology offers an effective approach to address the challenges of recycling and reusing retired batteries [6].

Unlike traditional battery systems with fixed series or parallel cell connections, RBS integrate power switching networks at the cell or

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module level. This architecture allows dynamic reconfiguration of the battery pack through real-time state monitoring and advanced control algorithms, as shown in Fig. 1. During operation, the electrical connections between cells can be adjusted instantly. This enables RBS to support a wide range of input and output voltages and to adapt to various load and energy source requirements. Compared with traditional fixed-topology systems, RBS can improve energy utilization and achieve effective cell balancing. This is realized by selectively controlling the operating duration of individual cells in different states, thereby maintaining pack-level uniformity. Moreover, precise control of the power switches, combined with accurate detection of faulty cells, enables the isolation of abnormal cells. This ensures stable system operation even in the presence of cell-level irregularities. This flexibility also supports the repurposing and cascaded use of retired batteries. As a result, RBS promotes a circular economy and enhances the overall sustainability of energy storage systems.

With the advancement of RBS technology, extensive research has been conducted on various aspects, including topology design and system control and operation. However, existing studies and reviews have largely focused on system management and control, with limited attention to application-specific analysis. Despite significant research on reconfigurable battery system (RBS) topologies and the proposal of various architectures, the applicable scenarios and performance differences among these topologies remain insufficiently characterized. Moreover, comprehensive evaluations and systematic comparisons of existing topologies are still lacking. By analyzing RBS topologies, it is possible to identify more efficient architectural designs. At the same time, exploring and analyzing efficient RBS architectures can advance battery technology, deepen the overall understanding of reconfigurable battery systems, and create additional opportunities for innovation in the energy sector.

In this paper, a comprehensive review and discussion of the design and operational characteristics of reconfigurable battery systems are provided. It analyzes the advantages and challenges of different topologies across various application areas, offering insights and guidance for future research and engineering applications. The rest of this paper is organized as follows. The existing RBS topology architecture is detailed in Section 2. Section 3 presents the operational characteristics of the RBS. Section 4 introduces the battery equivalent circuit model and analyzes the maximum circulating current generated during the RBS parallel process through the battery equivalent circuit. The reconfiguration methods of the RBS are thoroughly discussed in Section 5. Section 6 evaluates the existing RBS topologies based on relevant metrics and explores the applicable scenarios for these topologies. And section 7 concludes this paper.

2. Existing reconfigurable battery system topology

The RBS can reconfigure the arrangement of battery cells by controlling the adjustment switches, and exhibit a high level of fault tolerance with respect to open-circuit conditions and cell failures within the system [7]. Existing reconfigurable battery topologies are commonly classified according to the number of switches employed per cell into two-switch, three-switch, four-switch, and sub-two-switch topologies. To achieve the required controllability, power switching devices such as MOSFETs, IGBTs, and relays are typically employed as practical switching components for reconfiguring electrical connections among cells. In practical applications, MOSFETs are frequently adopted in reconfigurable battery systems because of their low switching losses and capability to support high-frequency operation [8,9]. Consequently, the analysis of switching components in this study focuses on MOSFETs. In most reconfigurable battery systems, the intrinsic body diode of a MOSFET allows unintended conduction in the reverse direction. To prevent bidirectional current flow where required, two MOSFETs are typically connected in reverse series (also known as back-to-back or common-source configuration) to block undesired current paths through their body diodes, as shown in Fig. 2. Under these conditions, the increased number of MOSFETs results in higher cost and additional power loss for switches configured with RS pairs. The choice of switching elements significantly affects the overall cost, power losses, and system reliability of reconfigurable battery systems. Therefore, clearly defining the switching configuration is essential for their design and operation. This section provides both a summary of existing topologies and a clarification of the corresponding switch configurations.

2.1. Sub-two-switch topology (self-X)

The sub-two-switch topology mainly refers to the self-X multi-cell battery pack architecture, which extends the traditional two-switch topology by using fewer than two switches per cell. This topology was introduced in [10–14], as shown in Fig. 3. It represents an extension of the series-connected topology. In addition to enabling series and bypass control of individual cells, it also allows parallel control of battery units. Compared with the series-connected topology, this architecture offers more flexible control and is suitable for more complex operating conditions. Here, self-X represents self-reconfiguration, self-balancing, self-optimization, and self-repair.

The Self-X battery architecture employs a switch array to control an $m \times n$ matrix of battery cells. Each cell is individually connected or disconnected through a single switch, forming a switch control unit. Within this structure, n switch control units are paralleled to increase current capacity, while m units are connected in series to raise the

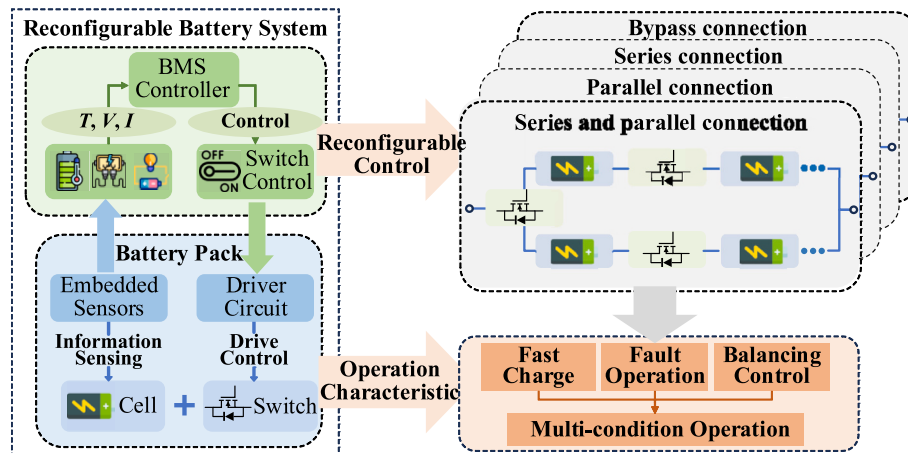


Fig. 1. Schematic diagram of reconfigurable battery system operation and architecture.

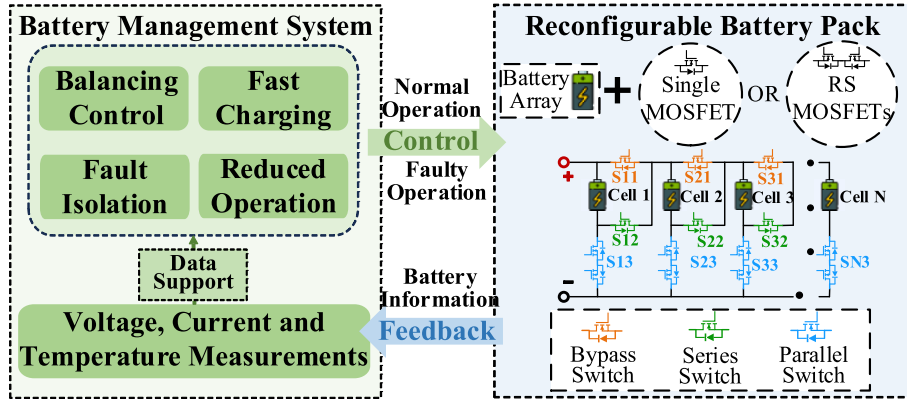


Fig. 2. Schematic diagram of a reconfigurable battery system and switch.

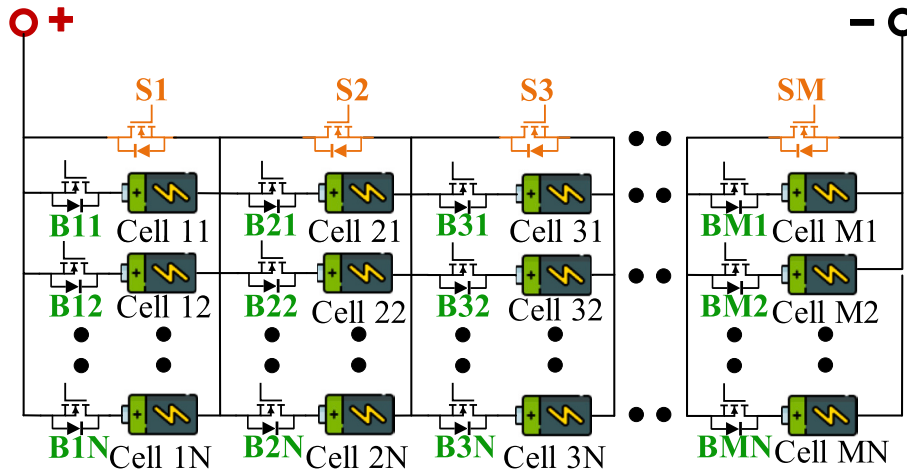


Fig. 3. Schematic diagram of self-X introduced in [10].

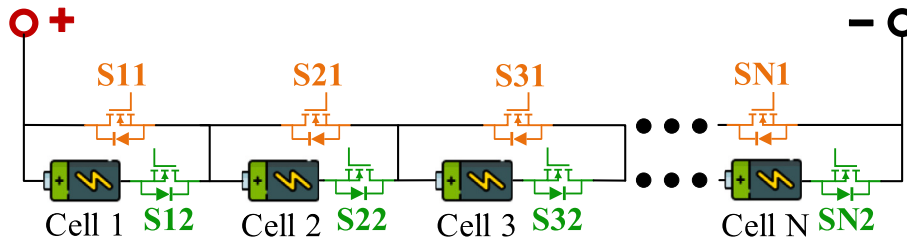


Fig. 4. Schematic diagram of series-connected battery topology introduced in [15].

output voltage. The bypass switch S_i ($i = 1, 2, 3, \dots, m$) controls the bypass of the corresponding parallel unit group. For example, when switch S_1 is closed, the corresponding controlled unit group $Cell_{1j}$ ($j = 1, 2, 3, \dots, n$) should be bypassed [14], meaning that the series-parallel switches B_{1j} ($j = 1, 2, 3, \dots, n$) need to be opened. The switches B_{1j} ($j = 1, 2, 3, \dots, n$) are series-parallel switches that simultaneously control the series and parallel connections of the corresponding battery cells. In this topology, the series and parallel connections of the batteries are implemented simultaneously, without the need for additional reconfiguration operations. Consequently, the failure of a single battery cell affects the output of the entire topology and simultaneously reduces the effective capacity of both the series and parallel connections within the system.

2.2. Two switches topology

The key characteristic of the two-switch topology is that it employs only two switches per battery cell. This makes it one of the simplest configurations among reconfigurable battery architectures. Due to the minimal number of switches and the straightforward system design, this topology offers advantages in terms of cost and complexity. However, its reconfiguration flexibility is significantly limited. It can only achieve series connection or bypass states for individual cells, and it does not support parallel connection between cells.

Currently, there are primarily two types of switch-based battery topologies: the series type and the half-bridge type. The series-connected architecture was introduced in [15–20], representing a simple reconfigurable topology formed by combining battery strings with switching elements. In practical applications, each switching position in this configuration typically requires only a single MOSFET. Among them,

switch S_{i1} ($i = 1, 2, 3, \dots, n$) is a bypass switch that allows the cell to be bypassed, switch S_{i2} ($i = 1, 2, 3, \dots, n$) is a series switch that controls the series connection of the cell.

Another topology, different from the series-connected type, which integrates a half-bridge circuit with two switches, was proposed in [21–24]. The structural topology is shown in Fig. 5(a). Similar to the series-connected topology, this half-bridge topology can only support the series and bypass two states of battery. Each half-bridge circuit consists primarily of two MOSFETs, S_{i1} and S_{i2} , connected in the same orientation. These devices control the bypass and series configurations of the corresponding cell, respectively.

Based on the work in [21], a two-switch topology integrating a half-bridge circuit with enhanced configurability was introduced in [25], as illustrated in Fig. 5(b). This topology is also constructed from a half-bridge circuit. However, in contrast to the configuration presented in [21], it enables parallel and series–parallel hybrid connections among batteries. Its configurability exceeds that of the previously described two-switch architectures. The proposed topology supports greater flexibility in cell arrangements while maintaining a comparatively low cost, thus providing an improved balance between reconfiguration capability and system cost.

2.3. Three switches topology

Compared with the two-switch topology, the three-switch reconfigurable battery topology employs a larger number of switching devices. This results in significantly greater reconfiguration flexibility. Although the increased number of switches leads to higher implementation cost, it substantially improves system adaptability and fault tolerance. Several

advanced variants of the three-switch topology have been proposed in the literature. Their main goal is to further enhance architectural flexibility and system reliability.

A three-switch topology with a series-connected architecture was proposed in [26,27], as illustrated in Fig. 6. This topology retains the functionalities of the series-connected architecture. However, it does not provide further optimization and is limited in achieving more versatile functional configurations. Compared with the series-connected architecture, it demonstrates an enhanced fault tolerance rate [26], although this improvement comes with increased architectural costs and power losses. Relative to the previously described three-switch topology, this configuration exhibits lower flexibility, supporting only series and bypass connection states. In [26,27], this topology was introduced to validate the effectiveness of the proposed RBS path planning method.

A topology in which every two cells are interconnected by three switches was introduced in [28–31], with the connection state of each cell controlled by these three switches, and its configuration is illustrated in Fig. 7. The three-switch topology represents an optimized extension of the series-connected architecture, providing enhanced configurability. In addition to series and bypass configurations, this topology also supports parallel and hybrid series–parallel arrangements. These cannot be achieved with the traditional series-connected architecture. In this design, the switches S_{i1} , S_{i2} , and S_{i3} serve as the parallel, series, and bypass switches, respectively. They enable control of the cell in parallel, series, and bypass configurations.

A three-switch battery topology with a half-bridge configuration was proposed in [32], as shown in Fig. 8. This topology employs three unidirectional MOSFETs to control each battery cell. It incorporates a half-bridge circuit in which two MOSFETs, oriented in the same

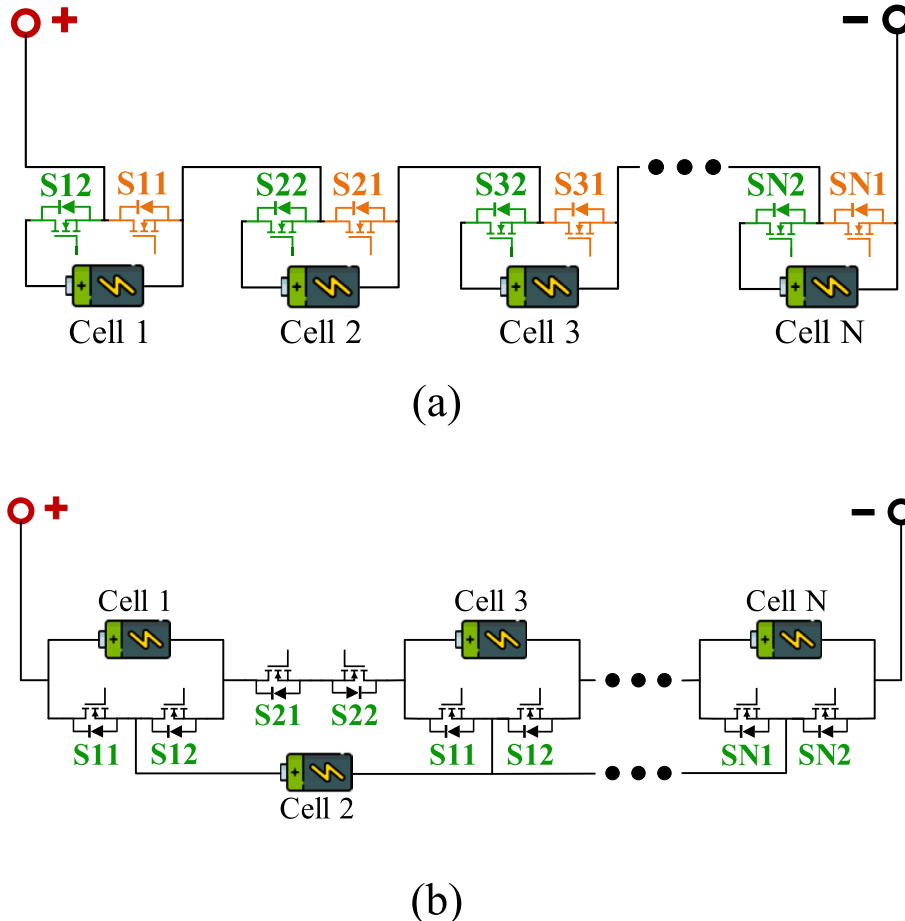


Fig. 5. Schematic diagram of half-bridge battery topology: (a) proposed in [21]. (b) proposed in [25].

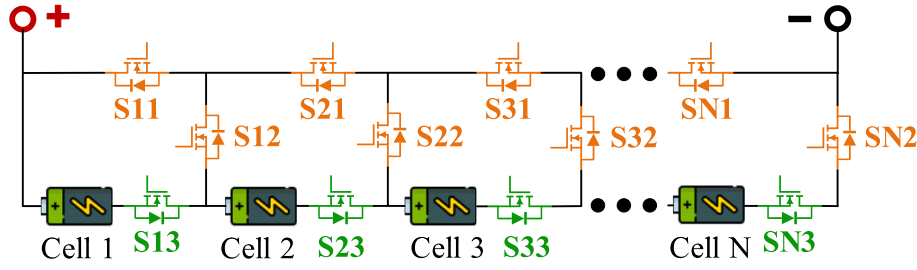


Fig. 6. Schematic diagram of a series-connected three-switch topology proposed in [26].

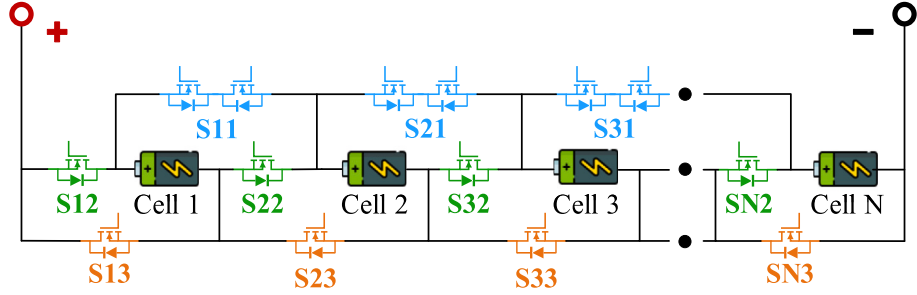


Fig. 7. Schematic diagram of the three-switch reconfigurable battery topology proposed in [28].

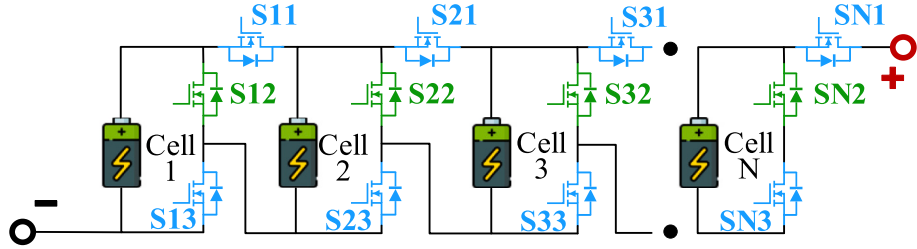


Fig. 8. Schematic diagram of the three-switch framework with half-bridge configuration proposed in [32].

direction, control the series and bypass configurations of the cell, respectively. Compared to the aforementioned three-switch topology, this topology has lower flexibility, as it can only implement the series, bypass, and parallel connections of cells. In the work presented in [32], this topology was primarily applied to optimizing the power systems of electric vehicles to enhance energy efficiency.

The improved three-switch topology based on [32] was introduced in [33–36], as illustrated in Fig. 9. In this topology, each cell is controlled by three switches, which simplifies implementation and control and maintains high flexibility. Each switch is implemented using three unidirectional MOSFETs. This enables the battery cells to be configured in series, parallel, bypass, or hybrid series–parallel modes. This architecture has also been adopted as the foundational framework for the Battery Modular Multilevel Management System (BM3) in [37–40].

A dependable, efficient, and scalable (DESA) three-switch architecture was proposed in [41], as illustrated in Fig. 10. This design

represents another innovative application of the half-bridge circuit within the three-switch topology. The proposed master–slave control method enables effective management of large-scale battery packs, allowing real-time monitoring and control of numerous cells and maintaining reconfigurability. In this architecture, the bypass switch is placed on the main line, while the parallel switch is located on the branch. This arrangement provides greater independence in parallel control. Compared with the half-bridge three-switch topology in [32], this design keeps the high flexibility and fault tolerance of the three-switch configuration. It also supports more comprehensive reconfiguration capabilities.

A three-switch topology, as illustrated in Fig. 11, was proposed in [42–44]. In this configuration, each cell forms an individual branch, with switches on the branches controlling the parallel and bypass connections of the cells. Compared with the topology in [33], where switches are placed on the main path, this design uses more switches. This leads to improved system reliability. The reconfiguration flexibility

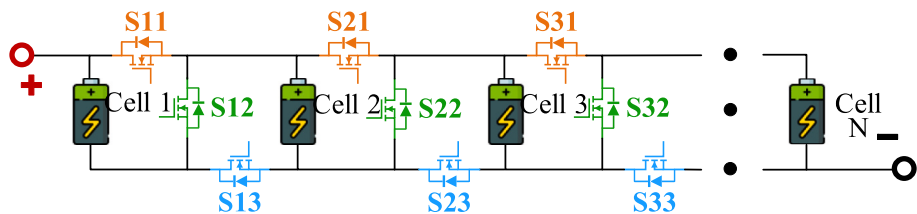


Fig. 9. Schematic diagram of the three-switch reconfigurable battery topology proposed in [33].

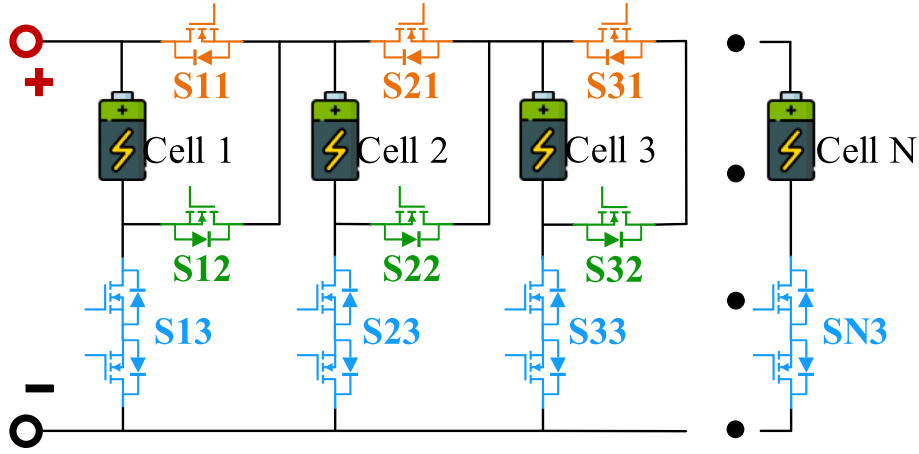


Fig. 10. Schematic diagram of the three-switch topology in DESA proposed in [41].

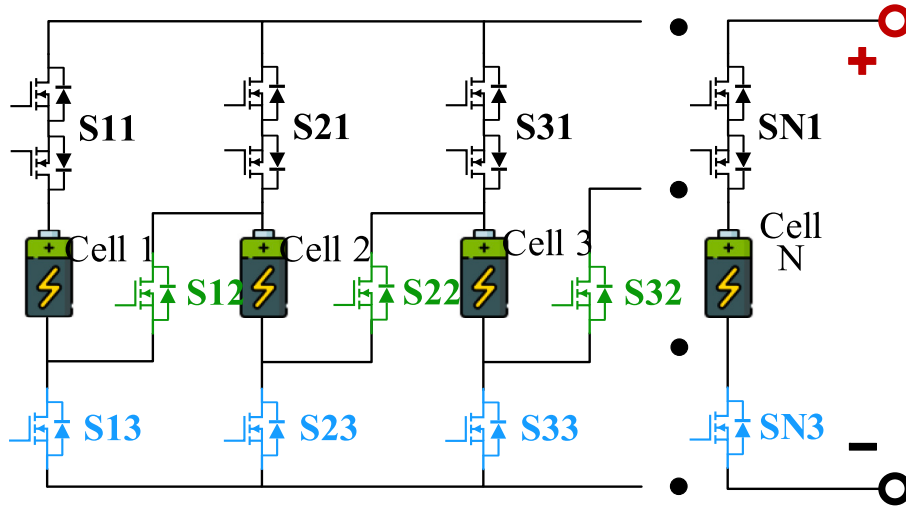


Fig. 11. Schematic diagram of the three-switch reconfigurable battery topology proposed in [42].

remains as high as in [33]. It supports four connection types: series, bypass, parallel, and series-parallel hybrid configurations. In this topology, switches S_{12} and S_{13} control the series and parallel connections, respectively. Bypassing a cell requires only opening switch S_{11} . No other switches need to be operated.

With the advancement of RBS architectures, dual-layer switch topologies have been gradually proposed. These topologies aim to better adapt RBS for large-scale energy storage applications. In this approach, individual cells in a single-layer topology are grouped into battery modules. Each module is itself built using a reconfigurable battery topology. The resulting structure, where both the cell layer and module layer adopt reconfigurable configurations, is referred to as the dual-layer RBS architecture. By controlling power electronic switches, this hierarchical design can flexibly adjust the connections between battery cells and between battery modules. This enables reconfigurable operation at both the cell level and the module level.

The currently proposed double-layer topologies all adopt three-switch configurations. Among them, the earliest three-switch double-layer topology was introduced in [8], as illustrated in Fig. 12(a). This topology employs different types of three-switch topologies (TTDS) for the inner and outer layers. The inner layer of the TTDS double-layer topology adopts the three-switch configuration proposed in [28], whereas the outer layer integrates the three-switch topology presented in [33]. Another double-layer topology, based entirely on the same type of three-switch topology (TTSS) for both inner and outer layers, was

introduced in [35,45,46], as shown in Fig. 12(b). In the TTSS double-layer topology, both layers adopt the three-switch configuration from [33]. Since the inner and outer layers of these dual-layer topologies are constructed from three-switch configurations, they retain the inherent flexibility of the three-switch design. In addition, as both layers utilize a reconfigurable architecture, the double-layer three-switch topology enables precise and efficient cell reconfiguration in large-scale energy storage systems. The effectiveness of this reconfiguration relies on the proper design and coordination of master-slave controllers for the inner and outer layers.

2.4. Four switches topology

The architectural development of the four-switch topology is similar to that of the three-switch topology, and it has already evolved into multiple structural forms. At the same time, the composition of the four-switch topology bears certain similarities to the three-switch topology. This also implies that the flexibility of the four-switch topology is comparable to that of the three-switch topology, allowing the control cell to switch between various connection states. However, the increase in the number of switches also entails higher hardware costs, which should be taken into account when selecting the RBS topology in practice.

Based on the three-switch topology proposed in [33], a four-switch topology can be obtained by adding a switch to each cell branch, as

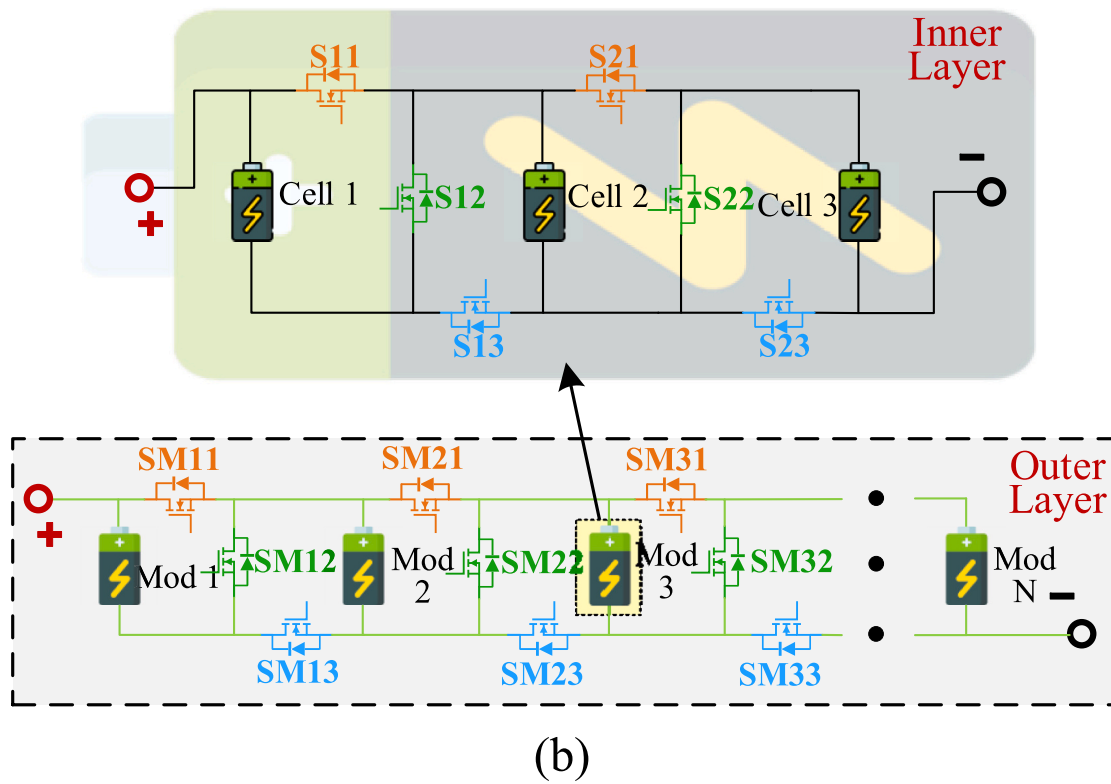
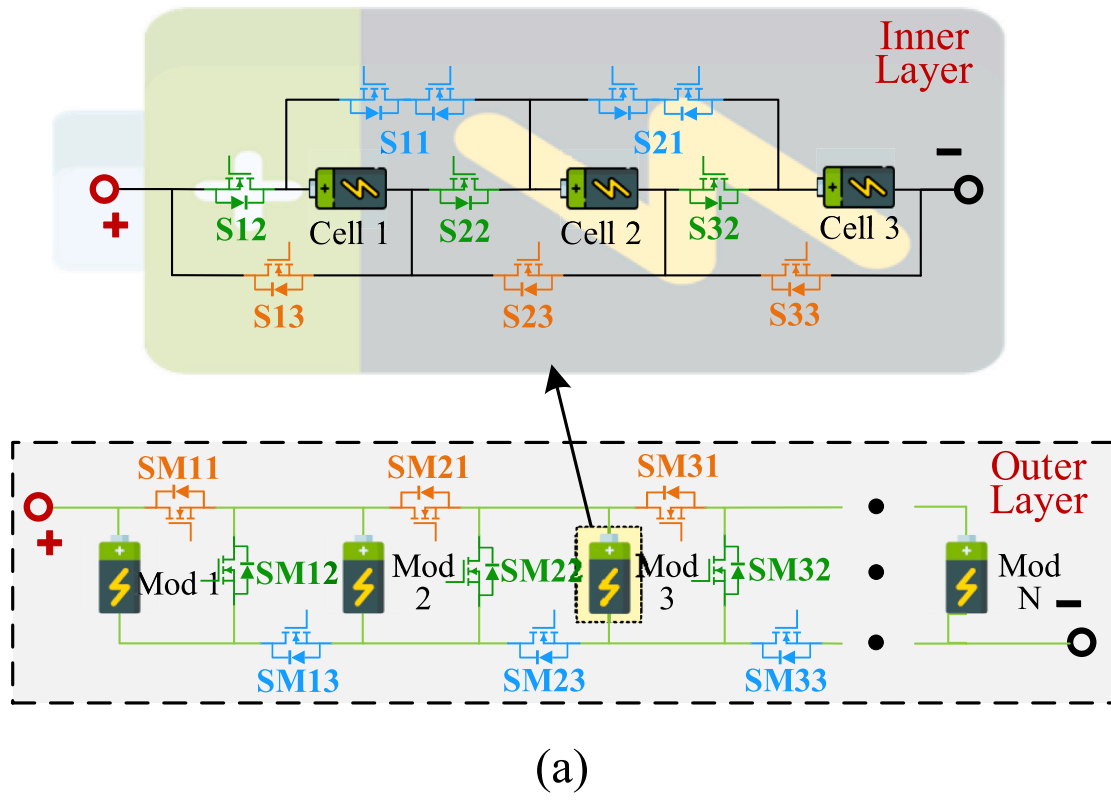


Fig. 12. Schematic diagram of double-layer RBS topology: (a) TTDS double-layer topology proposed in [8]. (b) TTSS double-layer topology proposed in [45].

illustrated in Fig. 13. This four-switch topology was proposed in [47,48], and the additional switch enables more precise control of the cell connection states. The topology provides high reconfiguration capability and strong fault tolerance. It supports series, bypass, parallel, and hybrid series–parallel configurations. Additionally, it enables precise isolation of faulty cells during the reconfiguration process.

By adjusting the series switch connection positions in the topology proposed in [47], so that the cell changes from the original main-line series to branch-series, a new four-switch topology as shown in Fig. 14 is obtained. This four-switch topology was introduced in [49–51]. This topology also possesses high reconfiguration capability and fault tolerance, enabling diverse reconfiguration configurations. Additionally, this topology has a high degree of cell independence, supporting completely independent cell control and easily handling cell anomalies during the reconfiguration process [49].

A JPL-type four-switch topology was used in [52–55]. Compared with the three-switch topology shown in Fig. 10, this four-switch topology adds a switch circuit between the two cells, as illustrated in Fig. 15. This topology inherits the flexibility of the four-switch configuration, supporting series, parallel, bypass, and hybrid series–parallel connections between cells. Moreover, the increased number of branches improves system reliability. It enables simpler and more effective bypassing of faulty cells during series, parallel, or hybrid configurations, thereby providing higher fault tolerance.

By adjusting the switch that connects two cells in the JPL-type four-switch topology to connect to the negative terminal of the next cell, the four-switch topology shown in Fig. 16 can be obtained, which has been used in [56]. This four-switch topology does not differ significantly from the JPL-type architecture, so its topology flexibility and fault tolerance are not changed. Only the corresponding switch states are adjusted when controlling the cell connection status.

Based on the topologies in [52,56], the proposed converter retains switches S_{12} and S_{13} to connect the positive and negative terminals of each module, resulting in the four-switch topology illustrated in Fig. 17 [57]. This topology retains the fault tolerance of the previously described four-switch design. However, its reconfiguration flexibility is significantly reduced. Series connections must be pre-established within each battery module, and the architecture does not allow control of series connections between different modules. Through switch operation, this configuration only supports bypass or parallel arrangements of the battery modules. A failure of a single cell within a module directly affects the entire module, resulting in lower reliability compared with the four-switch topology introduced earlier.

The half-bridge two-switch topology is expanded to a four-switch topology incorporating a cascaded H-bridge (CHB) [58–60], shown in Fig. 18. This architecture integrates the CHB multilevel converter, allowing direct DC-to-AC conversion for AC grids and enabling reconfigurable control of individual battery cells or single battery modules [58]. This four-switch topology with CHB differs from conventional RBS topologies that are directly connected to AC grids. In this design, closing switches in the same column simultaneously (for example, S_{11} and S_{13})

achieves bypass control of a cell, whereas closing switches in different columns simultaneously (for example, S_{11} and S_{14}) enables series control of a cell. Although direct parallel control of battery cells is not supported in this configuration, the topology integrates reconfigurable battery design with inverter functionality. This approach offers a valuable reference for the design and application of reconfigurable battery energy storage systems in AC grid environments.

Based on the four-switch topology proposed in [49] and the three-switch topology introduced in [33], a double-layer architecture can be constructed, with the inner layer formed by a four-switch topology and the outer layer by a three-switch topology (TFDS). This architecture was proposed in [61] and is illustrated in Fig. 19. The design retains the flexibility and reliability characteristics of double-layer topologies. However, it incurs higher costs, representing a limitation relative to the TTSS and TTDS topologies.

2.5. Five switches topology

To achieve flexible configuration among battery cells and meet the demands for high voltage and high capacity from the battery power supply, an early reconfigurable battery topology was introduced in [62–66]. This topology employs a larger number of switches, with each battery unit controlled by five switches, as illustrated in Fig. 20. The increased number of switches provides high reconfiguration flexibility and strong fault tolerance. It enables series, parallel, bypass, and hybrid series–parallel configurations, as well as a wide range of output voltages. However, the higher switch count leads to increased cost compared with previously presented reconfigurable topologies.

A five-switch topology, distinct from the aforementioned topologies, was proposed in [67]. This topology is an extension of the three-switch topology introduced in [28]. Compared with the three-switch topology in [28], an additional series switch is added to the negative terminal of each battery cell, and the parallel switch is divided into a set of switches connecting the positive and negative terminals of adjacent cells, as shown in Fig. 21. This five-switch topology inherits the flexibility and fault-tolerant capability of the three-switch topology. The increase in the number of switches also allows for more abundant reconfiguration possibilities than the three-switch topology, thereby enhancing its reliability.

2.6. Six switches topology

The six-switch topology is currently the topology with the largest number of switches used to control a single cell. An earlier six-switch topology was introduced in [68–70], as shown in Fig. 22. This topology is an extension based on the JPL-type four-switch topology. Unlike the four-switch topology, this six-switch topology has two additional switches connecting the terminals of adjacent cells. This is because the topology is designed with multiple input/output terminals, and the two added switches are primarily used for selecting the input/output terminals, allowing flexible switching of input/output ports. This provides

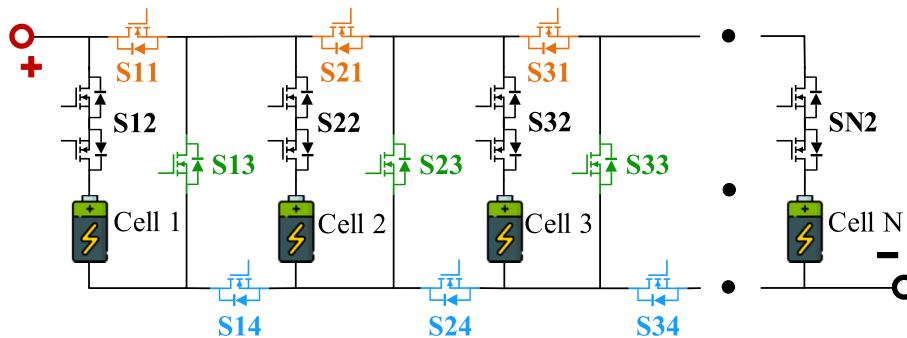


Fig. 13. Schematic diagram of the four-switch reconfigurable battery topology proposed in [47].

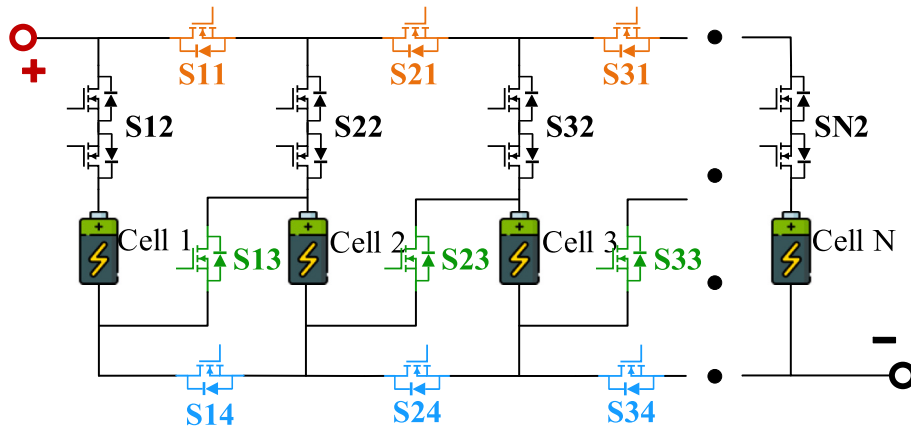


Fig. 14. Schematic diagram of the four-switch reconfigurable battery topology proposed in [49].

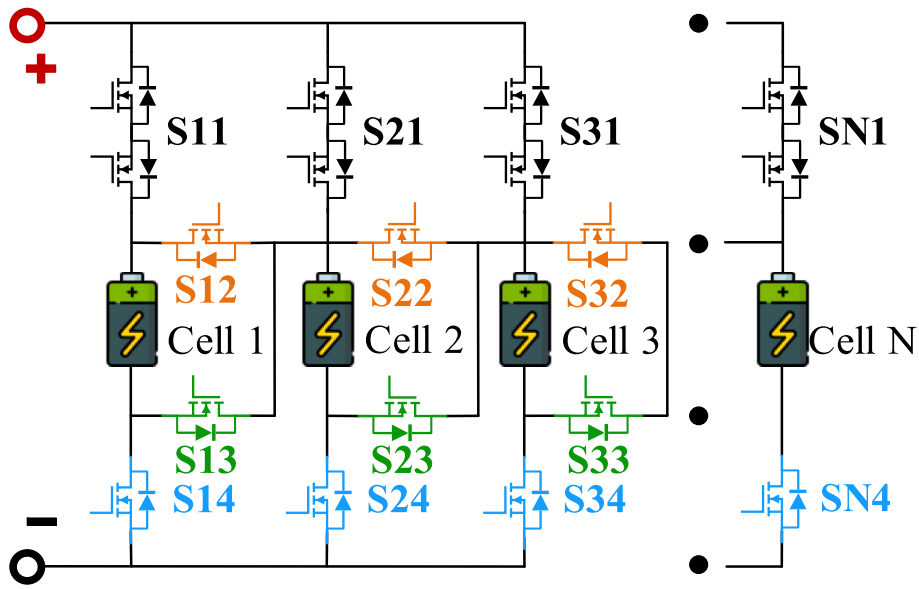


Fig. 15. Schematic diagram of the JPL-type four-switch reconfigurable battery topology used in [52].

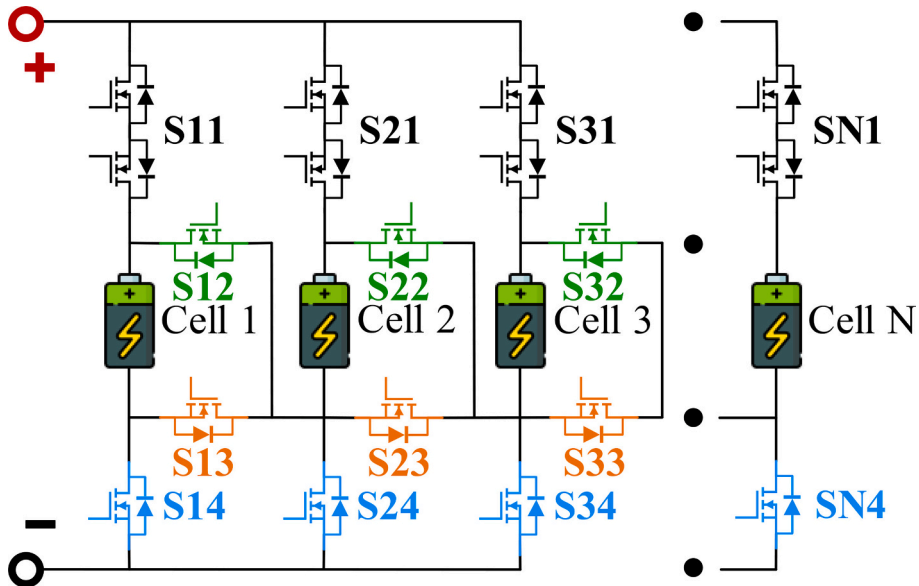


Fig. 16. Schematic diagram of the four-switch reconfigurable battery topology used in [56].

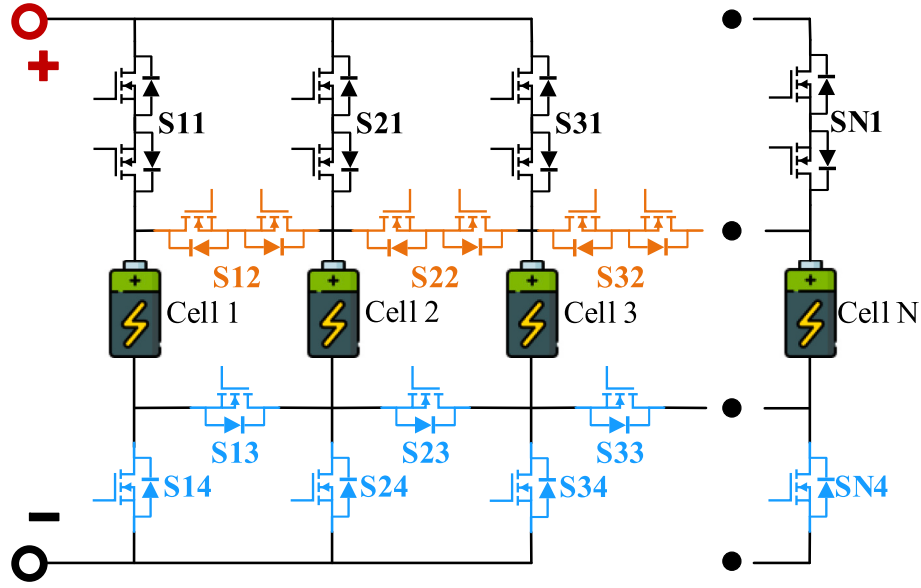


Fig. 17. Schematic diagram of the four-switch reconfigurable battery topology used in [57].

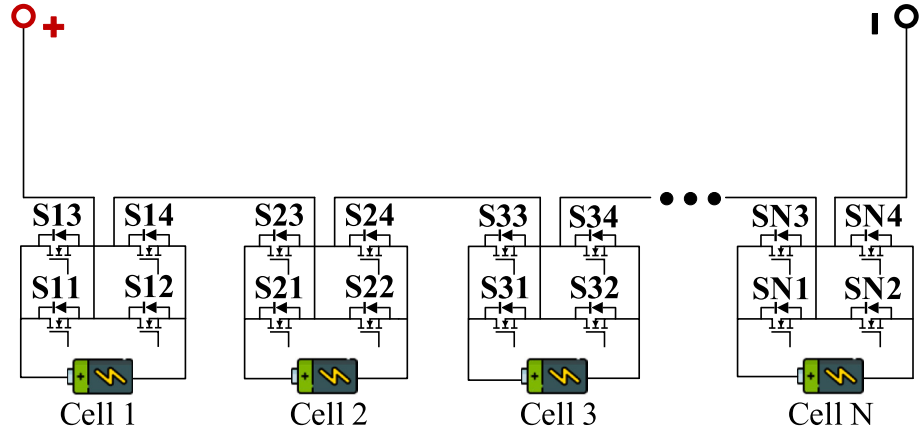


Fig. 18. Schematic diagram of the cascaded H-bridge four-switch reconfigurable battery topology introduced in [58].

the topology with high flexibility and fault tolerance.

2.7. Discussion

To evaluate the discussed topologies objectively, this section provides a brief comparison in Table 1 based on three criteria: control complexity, applicable energy storage scale, and suitability for second-life battery applications. The control complexity is assessed by considering both the hardware count and the control signal intricacy. Specifically, the hardware count is determined by the number of switches and sensors required, while the signal intricacy depends on the variety of connection modes supported by the topology. A five-level scoring system is used: a score of '1' indicates the lowest complexity (minimal switches and connection modes), whereas a score of '5' represents the highest. Regarding the energy storage scale, the assessment integrates cost, efficiency, reliability, and flexibility. Detailed discussions of these factors are provided in Section VII, with only the final results summarized here.

Furthermore, the tolerance to parameter heterogeneity is a key metric for evaluating whether a topology is suitable for second-life batteries. Given the significant inconsistencies in internal resistance and residual capacity among retired batteries, topologies that support only a single connection mode will suffer from a severe 'weakest link

effect', where the system's energy utilization is limited by the weakest cell. In contrast, topologies supporting diverse connections enable active power distribution. By dynamically reconfiguring the physical position and connection method of each battery within the topology, the degradation differences between cells can be compensated. Consequently, topologies with low degrees of freedom exhibit significant limitations in second-life battery applications due to their inability to decouple and control individual cell variations.

3. Operating characteristics of RBS

Unlike traditional fixed-topology battery systems, RBS offer dynamic reconfiguration capabilities. These capabilities improve energy efficiency, extend battery lifespan, and enhance fault tolerance. These advantages are a direct outcome of the distinctive operating characteristics of RBS. In essence, the operating characteristics largely determine the system's efficiency, reliability, and adaptability. A thorough understanding and analysis of these operating characteristics are therefore essential. They help reveal both the strengths and limitations of RBS and support the further development and practical application of this technology. This section provides an overview of the operating characteristics of RBS, focusing on their unique implementation approaches.

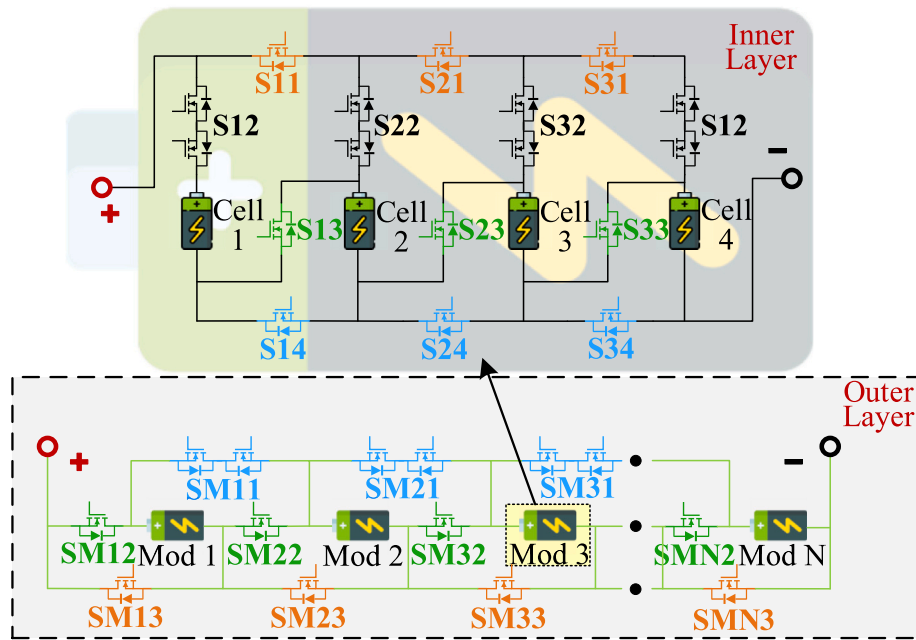


Fig. 19. Schematic diagram of TFDS double-layer RBS topology proposed in [61].

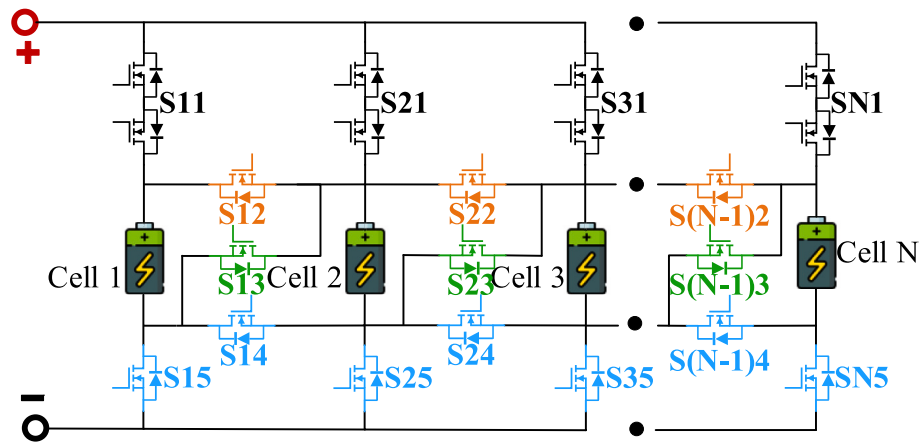


Fig. 20. Schematic diagram of early reconfigurable battery topology introduced in [62].

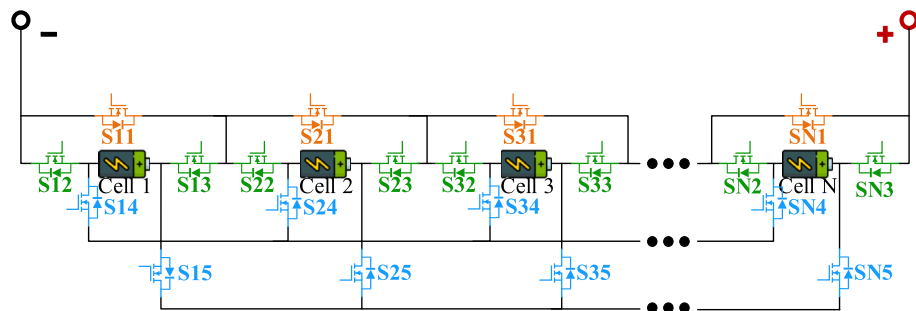


Fig. 21. Schematic diagram of the five-switch reconfigurable battery topology proposed in [67].

3.1. Balancing control

Battery imbalance remains a critical issue in both traditional battery systems and RBS. Cell-to-cell variations can significantly reduce system efficiency and increase safety risks, such as overheating or excessive heat generation. Effective balancing control is therefore essential to

ensure efficient and safe operation of battery systems. Balancing techniques are generally classified into two categories: active balancing and passive balancing. Currently, passive balancing remains the most widely adopted approach in practical battery systems.

For RBS, implementing balanced control extends beyond individual cell management and requires comprehensive oversight based on real-

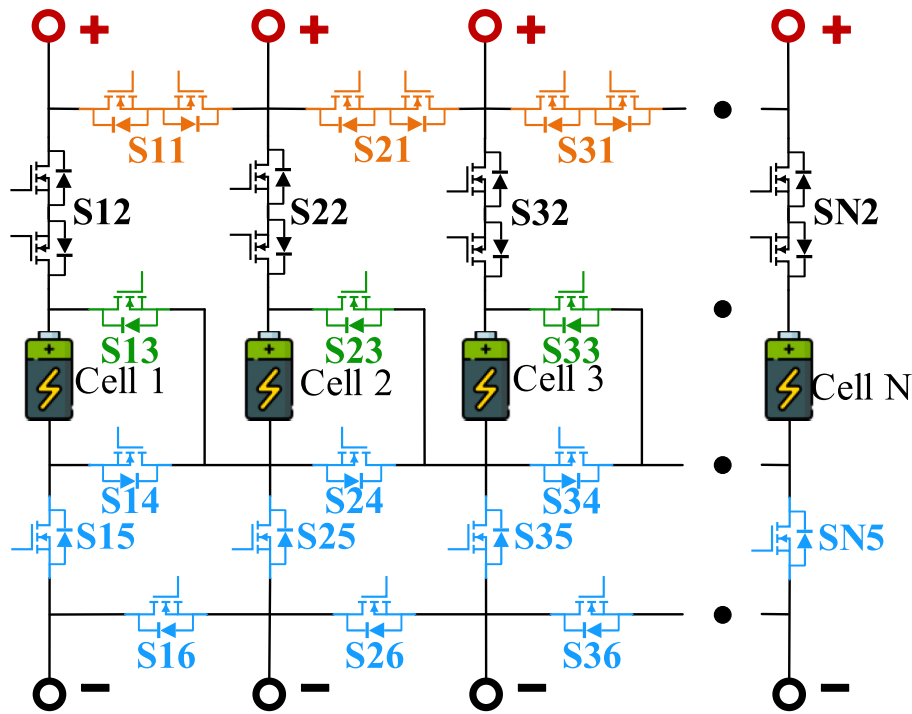


Fig. 22. Schematic diagram of early six-switch reconfigurable battery topology introduced in [68].

Table 1

Comparison of existing reconfigurable battery system topology designs (Y = yes, N = no, 1 = Low, 5 = High).

Design Diagram	Reference	Topology Type	Switches Per Cell	Series Connection	Parallel Connection	Series & Parallel Connection	Control Complexity (1–5)	Applicable Energy Storage Scale	Applicability of Retired Batteries	Valid Path (Flexibility)
Fig. 3	[10–14]	Self-X	≤ 2	Y	Y	Y	2	Large	Y	31
Fig. 4	[15–20]	Two	2	Y	N	N	1	Low	N	31
Fig. 5(a)	[21–24]	Switches	2	Y	N	N	1	Low	N	31
Fig. 5(b)	[25]		2	Y	Y	Y	3	Large	Y	31
Fig. 6	[26,27]		3	Y	N	N	2	Low	N	31
Fig. 7	[28–31]		3	Y	Y	Y	4	Large	Y	49
Fig. 8	[32]	Three	3	Y	Y	N	3	Medium	Y	31
Fig. 9	[33–36]	Switches	3	Y	Y	Y	4	Large	Y	49
Fig. 10	[41]		3	Y	Y	Y	4	Large	Y	49
Fig. 11	[42–44]		3	Y	Y	Y	4	Large	Y	49
Fig. 13	[47,48]		4	Y	Y	Y	4	Large	Y	49
Fig. 14	[49–51]		4	Y	Y	Y	4	Large	Y	49
Fig. 15	[52–55]	Four	4	Y	Y	Y	4	Large	Y	49
Fig. 16	[56]	Switches	4	Y	Y	Y	4	Large	Y	49
Fig. 17	[57]		4	N	Y	N	3	Low	N	49
Fig. 18	[58–60]		4	Y	N	N	3	Low	N	49
Fig. 20	[62–66]	Five	5	Y	Y	Y	5	Large	Y	113
Fig. 21	[67]	Switches	5	Y	Y	Y	5	Large	Y	113
Fig. 22	[68–70]	Six	6	Y	Y	Y	5	Large	Y	57

time parameter data for each cell or module. This control can be

Table 2

Comparison of cell balancing technologies.

Type	Scheme	Advantages	Disadvantages
Passive	Shunting Resistor	Cheap, Simple	Not effective, high energy loss
	Switched Capacitor	Simple, low voltage stress	Low equalization rate
Active	Single/Multi Inductor	Fast equalization speed	High current stress
	Windings	Fast equalization speed	High cost, high control complexity
	Transformer		

executed according to the balancing methods summarized in Table 2. Furthermore, RBS enables precise real-time adjustments at the cell level. As a result, inter-cell state balancing can be achieved using only the internal circuitry. No additional balancing circuits listed in Table 2 are required.

In current applications, RBS achieve balancing control through intelligent algorithms. These algorithms address voltage balancing, SOC balancing, and capacity balancing. The control strategy relies entirely on the system's internal circuitry to perform balancing. This approach ensures efficient utilization of electrical energy while keeping implementation costs under control. Regardless of the specific balancing method employed, the primary goal remains the same: to minimize differences among battery cells or modules. By doing so, it extends

system lifespan, reduces the occurrence of faults, and improves overall energy utilization efficiency.

3.2. Fault reconfiguration

Fault reconfiguration is a key technology that enables RBS to maintain efficient power output and reliable operation under fault conditions. It involves detecting performance degradation or faults in individual battery cells or modules and reconfiguring the system to preserve overall functionality and performance. The primary goal of fault reconfiguration is to minimize the impact of faults on system operation, thereby ensuring reliability and stability. Implementation relies mainly on real-time monitoring of battery states combined with fast switch control. This allows timely identification of faulty cells or modules. Once identified, the system isolates the affected components according to a predefined reconfiguration strategy, ensuring continued normal operation of the RBS. A well-designed fault reconfiguration mechanism enables the system to operate through faults, optimize the configuration to reduce energy losses, and ultimately improve overall availability and economic efficiency.

The purpose of fault reconfiguration is to ensure normal system operation and fulfill operational requirements. Therefore, the prerequisite for the system to perform fault reconfiguration is as follows:

$$V_{\text{Total}} - V_{\text{Fault}} = V_{\text{Work}} \quad (1)$$

This ensures that the system's fault reconfiguration is implemented under the condition of achieving the actual required voltage output, which is also a crucial condition for not affecting the normal operation of the system.

3.3. Parallel fast charging

In the charging state, RBS can connect multiple battery cells or modules in parallel. This configuration enables the delivery of a higher charging current for a short duration, thereby achieving a significantly elevated charging rate. For RBS, parallel fast charging distributes the current across multiple cells through multi-channel parallel paths. This approach reduces stress on individual cells, improves overall charging efficiency, and shortens the charging time [71]. However, the increased charging current may cause localized temperature rises within the cells, potentially compromising battery performance and safety. Therefore, in addition to enabling efficient fast charging, the system should integrate effective thermal management and appropriate safety mechanisms. These measures are essential to maintain safety and ensure long-term stability during high-rate charging.

During parallel fast charging, the primary concern is to prevent direct connection of batteries with mismatched SOC. Connecting unbalanced batteries in parallel can induce circulating currents, which may cause irreversible damage to the cells. Therefore, effective battery balancing control is essential during parallel fast-charging operation. Unbalanced batteries should not be directly paralleled. In practice, priority is typically given to charging batteries with lower SOC first. By extending the charging duration for low-SOC cells and shortening it for high-SOC cells, state balancing can be achieved simultaneously with the parallel fast-charging process.

3.4. Wide voltage range

RBS can dynamically adjust the series and parallel configurations of battery modules to regulate voltage under different operating conditions, ensuring efficient and stable operation of the battery system and satisfying load requirements. This capability also provides RBS with a wide voltage range of operation. A wide voltage range allows RBS to maintain stable performance across diverse voltage levels, thereby supporting various devices and load demands.

The specific range of input and output for RBS needs to be closely

analyzed in conjunction with the system architecture. The more flexible the architecture, the wider the range of input/output it can support. Among them, a two-switch topology with only series and bypass control capabilities can only support a wide-range operation in terms of voltage variation:

$$\min(V_{\text{battery}}) \leq V_{\text{in/out}} \leq \sum V_{\text{battery}} \quad (2)$$

This is because series connections do not alter the system current and only modify the system voltage. In contrast, for Self-X and multi-switch architectures with multiple control capabilities, such as parallel and series-parallel configurations, wide-range operation refers to substantial variations in both system voltage and current.

$$\min(I_{\text{battery}}) \leq I_{\text{in/out}} \leq \sum I_{\text{battery}} \quad (3)$$

The wide-range operating characteristics of the RBS enable it to achieve different voltage or current inputs and outputs solely based on its own architectural variations. This not only enhances the flexibility of the system but also improves energy utilization efficiency. Furthermore, this feature allows the RBS to be suitable for energy storage scenarios with varying load requirements and environmental changes.

4. System reconfiguration equivalent modeling

In RBS, an accurate battery model is not only the foundation for battery state estimation, but also a prerequisite for implementing functions such as system reconfiguration strategies, battery management, and fault diagnosis [72–74]. Therefore, constructing a high-precision and real-time battery model is of great significance for improving system performance and reliability. The equivalent circuit model (ECM) of lithium-ion batteries is currently the most widely used modeling method in RBS, primarily simulating the electrochemical characteristics of batteries through components such as resistors, capacitors, and voltage sources [75,76]. In existing studies, the battery equivalent circuit model commonly used in RBS is the first-order resistor-capacitor (RC) model [77–79]. Additionally, to better investigate the operational configuration of RBS, this section focuses on analyzing the reconfiguration of RBS batteries in conjunction with the battery equivalent circuit.

4.1. First-order RC model (Thevenin model)

The first-order RC model is also known as the Thevenin model. Compared with the Rint model, the Thevenin model adds an RC network [80,81], as shown in Fig. 23. The Thevenin model combines the advantages of high accuracy and low computational complexity, and is widely used in the fields of battery balancing and state estimation [26].

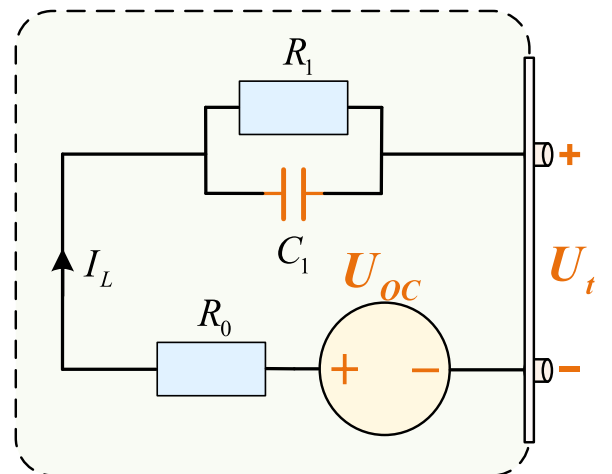


Fig. 23. The Thevenin Model and The First-order RC Model.

This model mainly consists of an open-circuit voltage source, an ohmic resistor, and an RC network. The RC network is composed of a polarization resistor and a polarization capacitor, and is mainly used to describe the polarization response of the battery. The electrical behavior of the Thevenin model can be expressed as follows:

$$\begin{cases} U_{1,k} = e^{-\Delta t/\tau_1} U_{1,k-1} + I_{L,k} R_1 (1 - e^{-\Delta t/\tau_1}) \\ U_{t,k} = U_{OC,k} - U_{1,k} - R_0 I_{L,k} \end{cases} \quad (4)$$

where U_1 represents the terminal voltage of the RC network, Δt is the sampling time, and τ_1 is the time constant describing the polarization characteristics of the battery, which satisfies: $\tau_1 = R_1 C_1$.

The Thevenin model simultaneously considers the ohmic polarization and electrochemical polarization of lithium-ion batteries. It can accurately describe the charging and discharging process of lithium-ion

batteries under constant temperature and constant current conditions and is currently the most commonly used ECM [74,82].

4.2. Series connection

Since the RBS is a battery system containing switching devices, the modeling process of the RBS should reflect the impact of the switching devices [83]. Generally, the influence of the switching devices on the circuit is primarily affected by the total resistance generated along the conductive path during their conduction [84,85]. Therefore, in this context, the modeling mainly considers the conduction resistance of the MOSFET, which is denoted as R_s^M in the diagram.

The series control of the RBS composed of M batteries is illustrated in Fig. 24. Since the system is in a series configuration at this stage, the

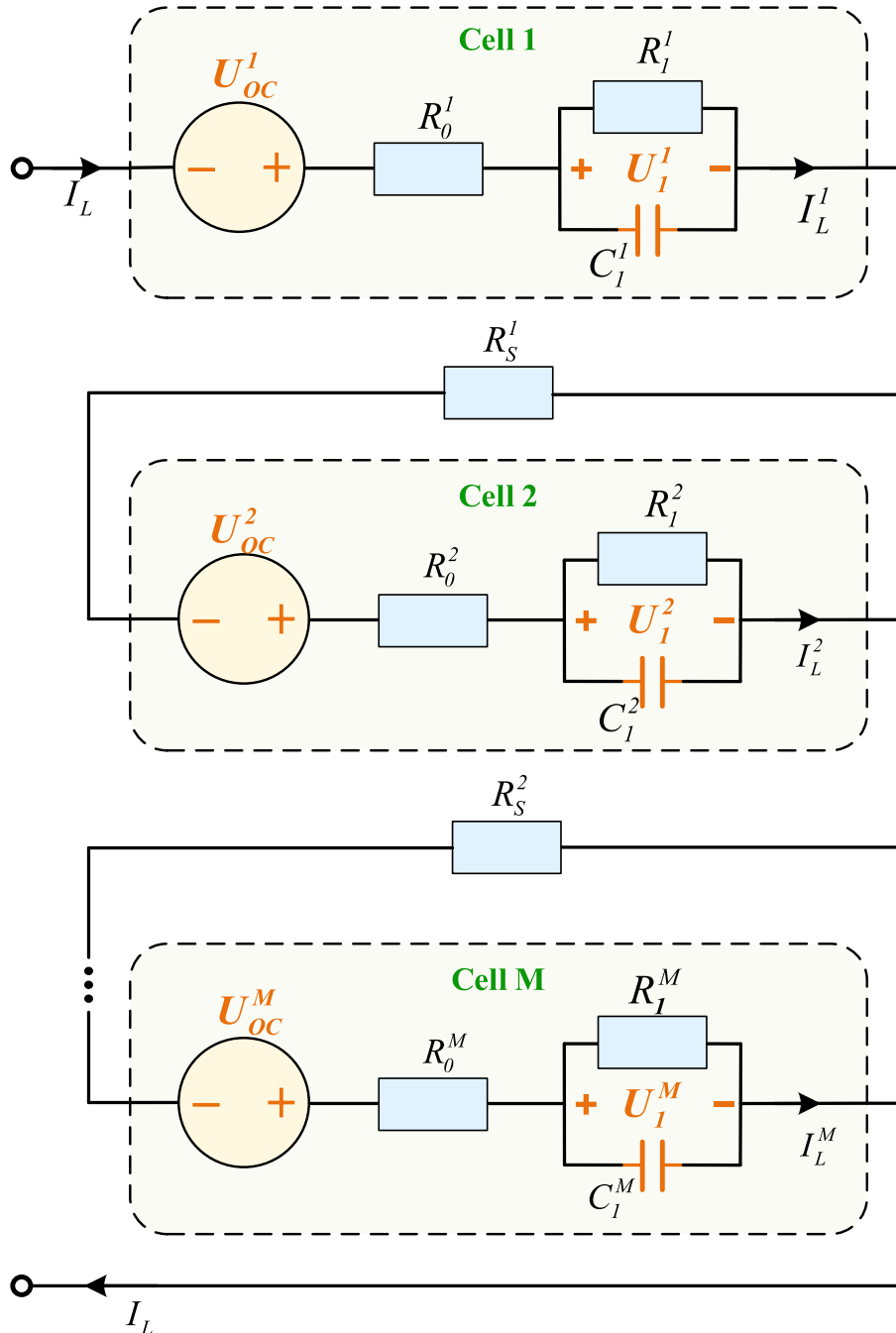


Fig. 24. Schematic diagram of ECM for series batteries.

total current of the system is equal to the current passing through each battery, that is, $I_L = I_L^1 = \dots = I_L^M$. At this time, the output voltage of the i -th battery U_t^i is given by:

$$U_t^i = U_{OC}^i - R_0^i I_L - U_1^i \quad (5)$$

where U_{OC}^i represents the open-circuit voltage of the i -th battery, and U_1^i represents the terminal voltage of the RC network in the ECM of the i -th battery. When the batteries are connected in series, the system voltage is primarily composed of the voltages of each battery along with the voltage generated by the internal resistance of the switch:

$$U_k = \sum_{i=0}^M U_t^i + \sum_{j=0}^{M-1} R_S^j I_L \quad (6)$$

In a battery string containing mismatched cells, the charge and discharge cutoff conditions of the entire string are determined by the cell that first reaches its limit. Specifically, during charging of the battery pack, the charging process is terminated when the cell with the highest voltage reaches its full-charge voltage. At that point, the voltage of each cell in the string is recorded as its respective cutoff voltage [86].

$$V_{\text{cut-charge}} = U_t^i + U_{\text{full}} - \max\{U_t^1, U_t^2, \dots, U_t^M\} \quad (7)$$

where U_{full} represents the voltage of a fully charged battery. In discharge mode, the cutoff voltage for each cell in the string corresponds to its voltage when the lowest-voltage cell reaches its fully discharged state.

$$V_{\text{cut-discharge}} = U_t^i - \min\{U_t^1, U_t^2, \dots, U_t^M\} + U_{\text{fdchar}} \quad (8)$$

where U_{fdchar} represents the battery voltage in a fully discharged state.

In series-connected battery strings, mismatched cells prevent

complete charging or discharging of the entire pack. This charging/discharging discrepancy significantly reduces the overall operational efficiency of the battery system. Furthermore, uneven charge and discharge processes accelerate battery degradation, thereby shortening the service life of the cells. Such imbalances also negatively affect the subsequent reconfiguration capabilities of the RBS.

4.3. Parallel connection

During the process of connecting batteries in parallel, the positive terminals or negative terminals of each battery need to be connected together [87]. In an RBS, this parallel connection operation should be carried out by switches that control the connection of the battery terminals, which are distinguished in the diagram as switches R_S^+ and R_S^- . The parallel control of an RBS composed of M batteries is illustrated in Fig. 25. At this time, the total current of the system is composed of the currents flowing through each battery in the system:

$$I_L = I_L^M + I_S^{M-1} = \sum_{i=0}^M I_L^i \quad (9)$$

where I_L^i is the current in the i -th battery ECM.

During the process of battery parallel connection, the most inevitable issue is the circulating current caused by voltage differences between the batteries. At the same time, this circulating current also increases the current flowing through the corresponding switch. Therefore, when selecting a switch, it is usually necessary to analyze the maximum circulating current between the batteries, as this affects the normal operation of the switch [88,89].

Assuming that all batteries and switches share the same internal resistance respectively, we have:

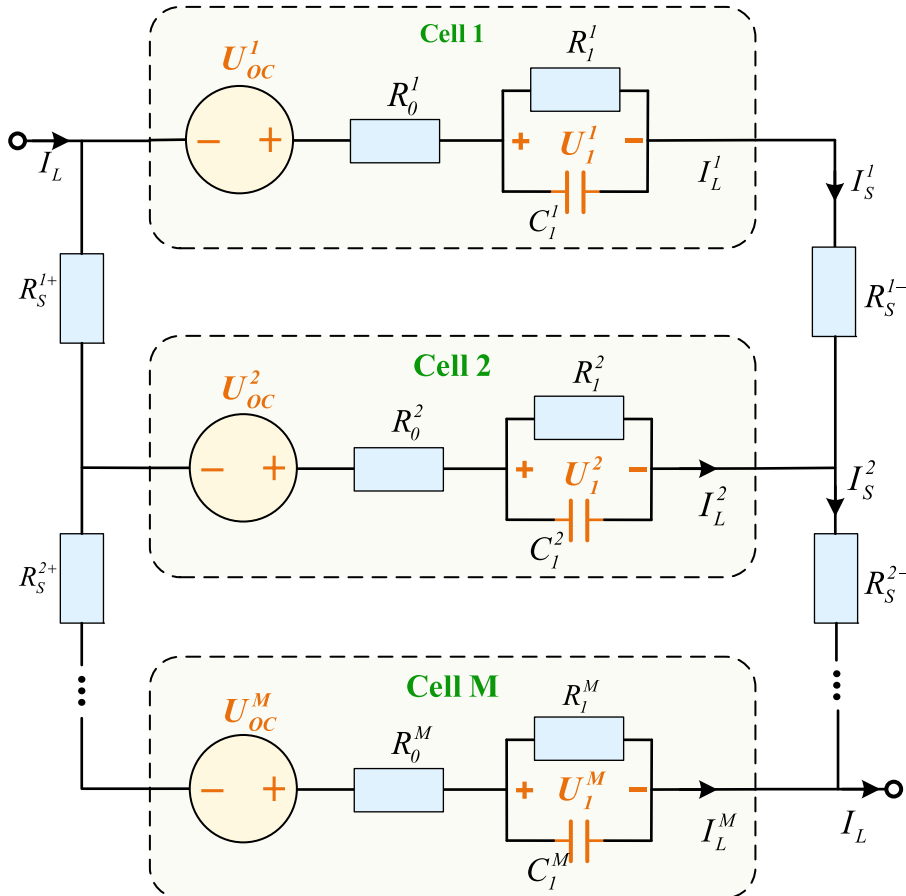


Fig. 25. Schematic diagram of ECM for parallel batteries.

$$R_0^u = R_0, R_S^{u+} = R_S^{u-} = R_S, u = 1, 2, \dots, M \quad (10)$$

At this moment, the internal resistance ratio between the switch and the battery is:

$$\lambda = R_S/R_0 \quad (11)$$

On this basis, the equivalent circuit parameters of an RBS consisting of K batteries in a parallel configuration can be derived as:

$$V_c^K = \begin{cases} V_{OC}^1, & K = 1 \\ \sum_{k=1}^{K-1} \left(\prod_{j=k+1}^K (1 - \alpha_j) \alpha_k V_{OC}^k \right) + \alpha_K V_{OC}^K, & K > 1 \end{cases} \quad (12)$$

$$\alpha_k = 1 - \frac{1}{\alpha_{k-1} + 2\lambda + 1} \quad (13)$$

Based on the equivalent circuit under the parallel connection of two batteries, and by combining Eq. 12 and Eq. 13, the circulating current between the two batteries can be expressed as:

$$I_S^m = \frac{|V_c^m - V_c^{M-m}|}{(\alpha_m + \alpha_{M-m} + 2\lambda)R_0} \quad (14)$$

The rated current of the switch should be determined by setting the lower limit of the switch carrying current value I_S^r according to the maximum value $I_S^{\max}$ of the circulating current generated under operating conditions:

$$I_S^{\max} \leq I_S^r = \frac{V_{OC}^u - V_{OC}^l}{\min_{1 \leq m < M} (\alpha_m + \alpha_{M-m})R_0 + 2R_S} \quad (15)$$

where V_{OC}^u and V_{OC}^l represent the upper and lower limits specified by the system for the battery OCV, respectively.

Due to cell imbalance, the circulating current generated during parallel operation can significantly increase battery temperature and substantially reduce operational efficiency during parallel charging or discharging in RBS. Therefore, effective cell balancing must be performed before initiating parallel connection of the batteries. Furthermore, the maximum circulating current observed in parallel operation corresponds to the peak balancing current that occurs during the system's active balancing process.

5. Reconfiguration method

The operation of RBS fundamentally relies on battery reconfiguration. By controlling this reconfiguration process, RBS can dynamically adjust the connection patterns among battery cells. This enables effective management of the overall input and output characteristics of the system [90]. However, since batteries are power sources, reconfiguration must carefully avoid critical issues. These include short circuits, voltage mismatches during parallel connection, and faulty cells being connected in series with healthy ones [91,92]. Such considerations require thorough evaluation of current and voltage constraints. Moreover, the choice of reconfiguration strategy under any topology is not determined only by these electrical and safety limits. It must also incorporate the specific characteristics of the adopted architecture and the actual operational requirements [93,94]. Ultimately, the primary objective of reconfiguration is to identify an optimal connection scheme. This scheme should minimize battery balancing time and cell degradation while satisfying the required input and output specifications [95].

5.1. Limiting factor

The reconfiguration method for RBS is a control strategy that enables rational and efficient energy distribution, allowing simultaneous charging or discharging of all batteries [96]. Since voltage and current are critical parameters for characterizing battery electrical behavior, the

design of RBS reconfiguration strategies must give full consideration to these two fundamental electrical constraints. Furthermore, RBS achieves diverse output voltage levels and system capacity requirements through precise control of individual battery cells [97]. This capability ensures compliance with stringent operating limits on current and voltage values [98,99].

5.1.1. Voltage limiting

During operation, the output voltage of the RBS must satisfy the load requirements. This means that the achievable voltage range after system reconfiguration should fully cover the operating voltage range demanded by the load. This is a key condition for determining whether the RBS can operate normally. At the same time, when controlling the connection status of batteries within the system, it is important to note that:

- 1) In a series configuration, any battery with a large voltage difference can cause overcharging or overdischarging [100]. Therefore, during the reconstruction control of the RBS, batteries with large voltage differences should not be connected in series into the system.
- 2) In a parallel configuration, any battery with a significant voltage difference can lead to accelerated deterioration of battery performance. At the same time, during parallel charging, it can also cause imbalance in the system's battery charging [101]. Therefore, during the reconstruction control of the RBS, the voltages of the batteries involved in parallel should be similar.

5.1.2. Current limiting

Each battery or battery module has a maximum charge/discharge current limit. Therefore, during the reconstruction process of the RBS, it is necessary to ensure that the current flowing through each battery or battery module does not exceed its rated current range.

During RBS operation, unequal current distribution among batteries is another critical issue that requires careful attention. Batteries with significant individual differences are prone to current overload. This leads to elevated cell temperatures, accelerated degradation, reduced service life, and in severe cases, irreversible damage to battery performance [102]. Therefore, when designing the reconfiguration strategy for RBS, an appropriate current allocation mechanism must be incorporated. This mechanism should ensure uniform current distribution across all connected batteries.

5.2. Normal operation

During normal operation, the reconfiguration strategy of RBS primarily ensures balanced charging and discharging of the batteries while satisfying the actual charge/discharge requirements of the system [103]. Under charging or discharging conditions, the system dynamically adjusts the series-parallel connection states of the batteries to meet the voltage and current demands of the load. Strictly speaking, RBS reconfiguration control involves precise regulation of the system's input/output voltage and current. To achieve efficient reconfiguration, the system typically incorporates redundant cells or battery modules. This design enhances balanced control, better meets operational requirements, and improves robustness against cell or module failures [13]. In the literature, various reconfiguration control methods have been proposed and experimentally validated to enable efficient RBS operation. Currently, strategies that integrate intelligent optimization algorithms represent the dominant trend in RBS reconfiguration research.

As shown in Fig. 26, one major category of reconfiguration methods focuses on battery balancing. These methods typically employ optimization algorithms, such as wear-leveling strategies or deferred convex optimization, to adaptively adjust the operating duration of individual batteries. This approach effectively minimizes total energy consumption during charging and discharging processes [103–105]. Another category adopts real-time path-planning techniques, often based on short-path

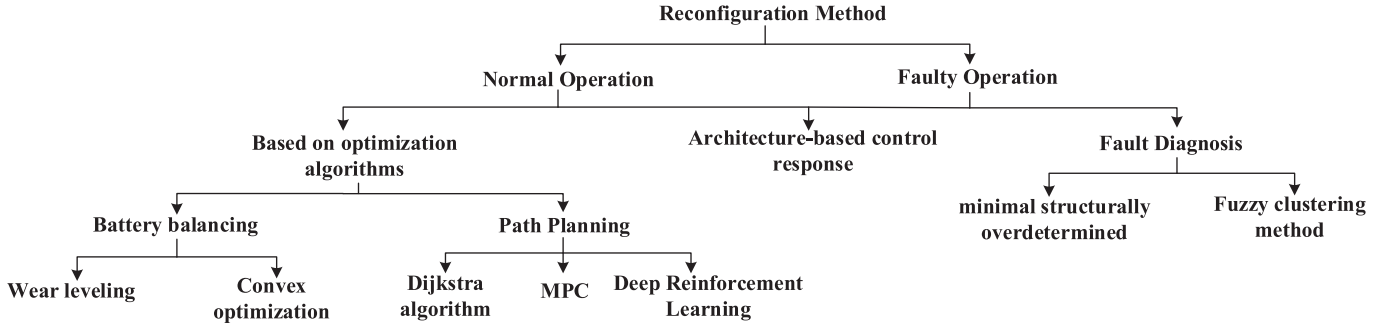


Fig. 26. Classification of refactoring methods.

planning or deep reinforcement learning. By performing goal-weight analysis, these methods dynamically search for the optimal system operating path, simultaneously achieving efficient system operation and battery balancing objectives [42,99,106]. Although the two types of reconfiguration approaches differ in their implementation, both effectively maintain battery state balance while ensuring normal system operation. As a result, they extend system runtime, reduce battery degradation, and sustain long-term high-efficiency performance [107].

5.2.1. Charging status

RBS can be charged in either series or parallel configurations. However, series charging is generally impractical due to the difficulty in

achieving voltage balance across cells in the series string during the charging process. In contrast, parallel charging allows batteries to naturally equalize their potentials. As a result, parallel charging is the most commonly adopted method for system-level charging [50]. In the parallel charging mode, the system's minimum termination voltage is determined by the charging cutoff voltage of the individual batteries [99]. During this process, charging priority is typically given to low-voltage cells. Once batteries reach the same potential (as illustrated in Fig. 27), balance is maintained by reducing the charging time of higher-voltage cells [50,108]. Parallel charging also enables the highest charging rate for the battery pack, thereby achieving fast-charging performance. To improve the efficiency and reliability of the discharge

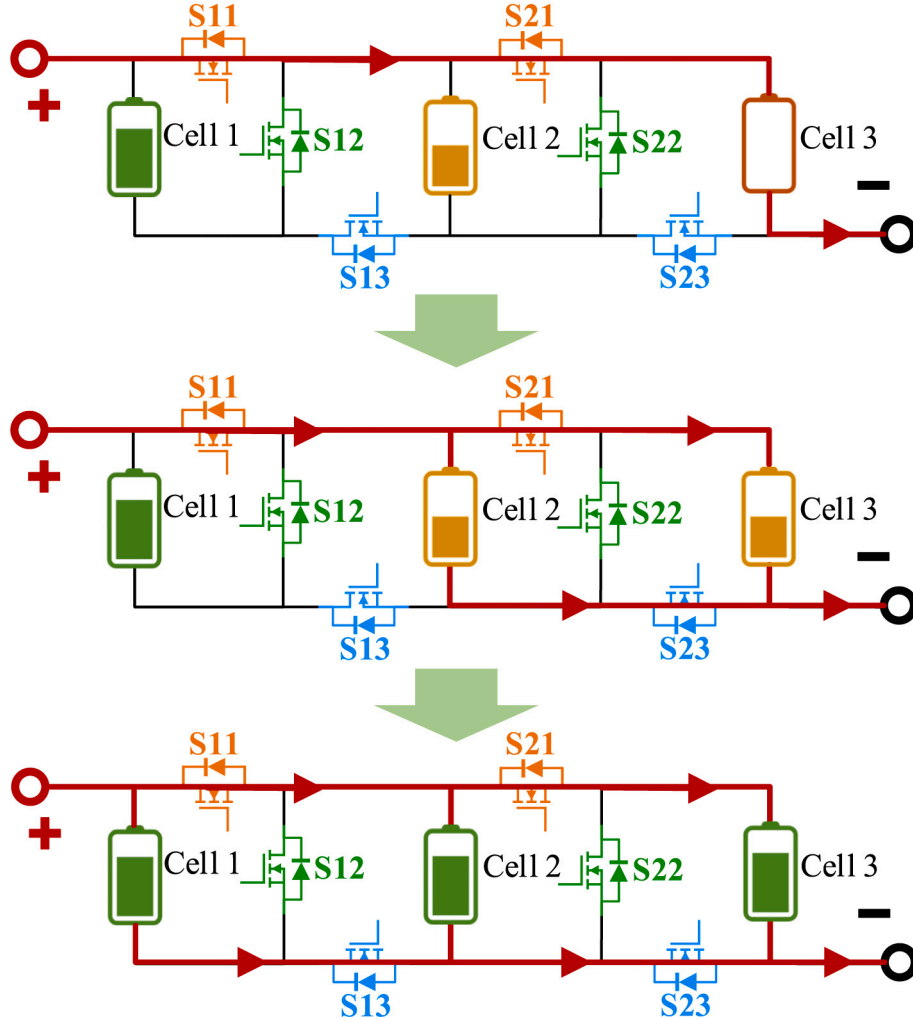


Fig. 27. Schematic diagram of the system charging status using a three-switch topology as an example.

process, model predictive control (MPC) is often employed to generate an optimal current trajectory. This approach allows precise targeting of desired temperature and voltage conditions, resulting in effective homogenization of battery SOC and temperature. The effectiveness of this discharge optimization strategy has been experimentally validated in previous studies [109–111].

5.2.2. Discharging status

RBS can achieve high current output through parallel discharge and high voltage output through series discharge [50]. The connection state of the RBS is primarily determined by the load requirements. A mixed series–parallel reconfigurable topology enables the system to accommodate a wider range of load conditions. Under heavy load conditions, the system is configured in series. In this mode, the output voltage depends on the number of batteries connected in series, with a larger number of batteries resulting in higher output voltage [112,113], as illustrated in Fig. 28a. Under light load conditions, the system switches to a parallel configuration to maintain stable operation and high efficiency across the full load range. In this case, the output voltage is limited to that of a single battery, while the current delivery capacity is significantly increased [113], as shown in Fig. 28b. During discharge, batteries with higher capacity or SOC are preferentially scheduled to discharge for longer durations [114]. This strategy is the reverse of the balancing approach typically applied during charging.

5.3. Faulty operation

The reliable operation of RBS under fault conditions depends critically on accurate fault detection. This requires precise monitoring of battery voltage, current, and temperature. A battery is typically classified as faulty when it exhibits abnormally high internal resistance, voltage values that are excessively high or low, or temperatures significantly above normal levels [115]. The system must strictly prevent faulty batteries from participating in any charging or discharging processes [108]. Upon detection of a fault, the primary response is to isolate

the faulty battery from the rest of the circuit. This is achieved by opening the switches associated with the faulty unit in series and parallel paths and closing the corresponding bypass switches. Subsequently, redundant healthy batteries are activated according to the current charge/discharge requirements. This ensures uninterrupted input/output performance of the system despite the presence of the faulty cell [104,116], as shown in Fig. 29. This fault-handling capability constitutes a major advantage of RBS over conventional battery systems. It significantly enhances fault tolerance and overall system reliability.

6. System management and control

The control of RBS is primarily managed by the BMS, which serves as the core component for monitoring and controlling battery operation. In traditional battery systems, the BMS mainly performs passive monitoring and protection functions. In RBS, however, its role has evolved significantly to include dynamic scheduling and reconfiguration of the battery topology. The specific control architecture of the BMS directly influences the system's response speed and the feasibility of implementing complex control algorithms. Currently, two dominant BMS architectures are used in RBS: centralized and distributed architectures [117,118], as shown in Table 3. In a centralized architecture, a single central controller collects global battery cell data and generates all switching commands in a unified manner. As the number of battery cells increases, this approach leads to a rapidly growing computational load on the central controller. Moreover, it introduces challenges related to complex wiring and single-point-of-failure risks, which become particularly pronounced in large-scale RBS applications. In contrast, the distributed architecture delegates control logic to module-level controllers, as shown in Fig. 30. By leveraging local autonomy, this design significantly improves system modularity, scalability, and fault tolerance [119,120]. The enhanced flexibility and local decision-making capability make distributed architectures particularly suitable for large-scale battery systems. Nevertheless, they impose stricter requirements on real-time coordination and communication consistency

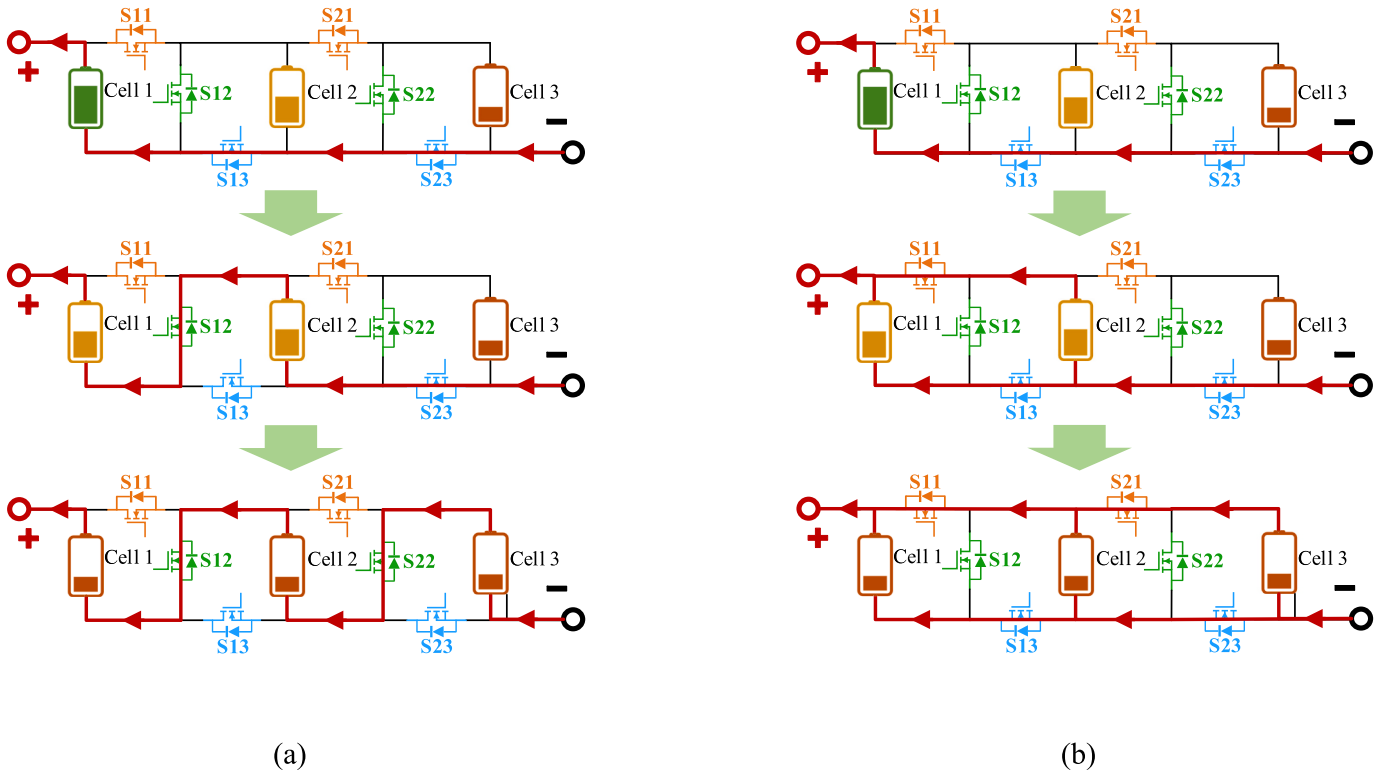


Fig. 28. Schematic diagram of the system discharging status using a three-switch topology as an example: (a) Heavy load. (b) Light load.

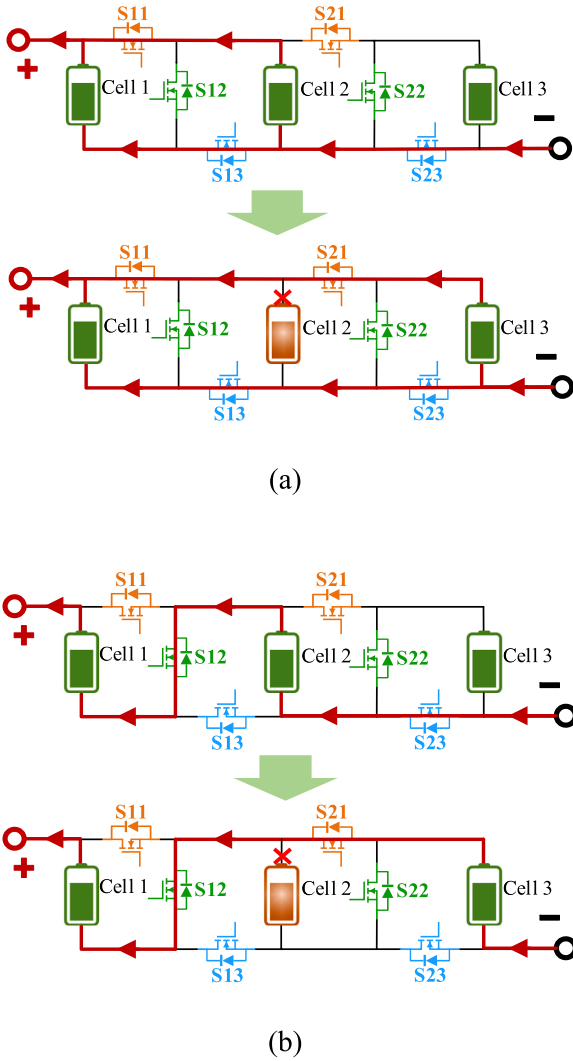


Fig. 29. Schematic diagram of faulty operation during discharge using a three-switch topology as an example: (a) Parallel operation. (b) Series operation.

Table 3

Comparison between centralized and distributed architectures.

Indicator	Centralized Architecture	Distributed Architecture
Control Core	Central controller unified control	The control logic can be delegated to module-level controllers.
Wiring Complexity	Complex	Lower
Architectural Reliability	Weak	Strong
Scalability	Poor	Excellent
Communication Requirements	Simple	Strict and complex
Potential Risk	Scale limitations and computational bottlenecks	Instantaneous topological switching may generate circulating current surges

across multiple nodes. This is essential to prevent circulating current surges resulting from uncoordinated topology switching.

6.1. Communication architecture

Within a BMS, the communication architecture functions as the critical interface bridging control logic and physical switch execution. It is imperative that this architecture maintains deterministic transmission

latency, particularly in environments characterized by severe electromagnetic interference (EMI). In practical deployments, the daisy-chain topology has emerged as the predominant physical layer solution for RBS due to its reduced wiring complexity and superior electrical isolation [121]. By employing differential signal transmission via capacitive or transformer coupling, the daisy-chain configuration effectively suppresses high-frequency common-mode noise originating from high-voltage buses, thereby ensuring data integrity despite kilovolt-level voltage offsets [122]. Nevertheless, the inherent serial architecture of the daisy-chain configuration constrains the effective utilization of bus bandwidth during high-frequency dynamic topology reconfigurations. Moreover, the vulnerability to single-point-of-failure risks remains a critical concern, as physical impairment of any individual node can precipitate a complete communication collapse for all downstream modules.

To enhance the robustness and data throughput of large-scale systems, fieldbus technology based on the CAN-FD protocol has been widely integrated into reconfiguration systems [123]. By extending the payload length and increasing the data-segment bit rate, CAN-FD effectively reduces the bus load during the concurrence of high-frequency sampled data and switching commands. This mitigation of bus traffic helps suppress control jitter caused by channel congestion. Furthermore, wireless BMS (wBMS) is emerging as a prominent research frontier for distributed systems requiring high power density and modularity [124,125]. By eliminating physical wiring harnesses, wBMS inherently resolves issues such as high-voltage isolation, harness aging, and vibration-induced contact failure, thereby streamlining mechanical design. Nevertheless, maintaining deterministic timing scheduling amidst multipath fading and interference remains a critical technical bottleneck for secure RBS reconfiguration. This evolution from simple physical connectivity toward deterministic networked collaboration not only ensures precise synchronization of control logic across nodes but also establishes the physical framework for implementing complex real-time decision-making algorithms.

6.2. Reconfiguration and real-time decision making

The core value of RBS lies in fundamentally addressing internal cell inconsistency through dynamic topological adjustments. However, as the number of controlled units n increases, the system encounters an exponential expansion of the switching state space. This poses a significant challenge for maintaining efficient and stable control. To enable high-performance decision-making within limited computational constraints, research has shifted from simple logic switching to sophisticated software-driven algorithms. Overall, current RBS control methods aim to identify the optimal configuration to minimize power consumption while maximizing capacity and lifespan within millisecond-scale intervals.

Real-time optimization strategies represent the most rigorous control methods in early RBS research. The core logic involves modeling topological reconfiguration as a constrained mathematical programming problem. Specifically, Mixed-Integer Linear Programming (MILP) is widely employed to describe the discrete switching states by solving objective functions within each decision cycle [126,127]. To further capture the dynamic characteristics of battery systems, Model Predictive Control (MPC) has emerged as a representative approach [84,94]. By integrating ECM, MPC predicts the evolution of the State of Charge (SOC) and temperature rise over a future horizon. This enables proactive topological pre-scheduling [110]. Despite their strong mathematical completeness, the high computational cost in large-scale systems remains the primary bottleneck for practical engineering applications.

To address the challenges of battery high non-linearity and the difficulty of acquiring precise physical models under complex operating conditions, data-driven methods based on Reinforcement Learning (RL) have demonstrated significant potential. Specifically, Deep Reinforcement Learning (DRL) learns optimal control laws through continuous

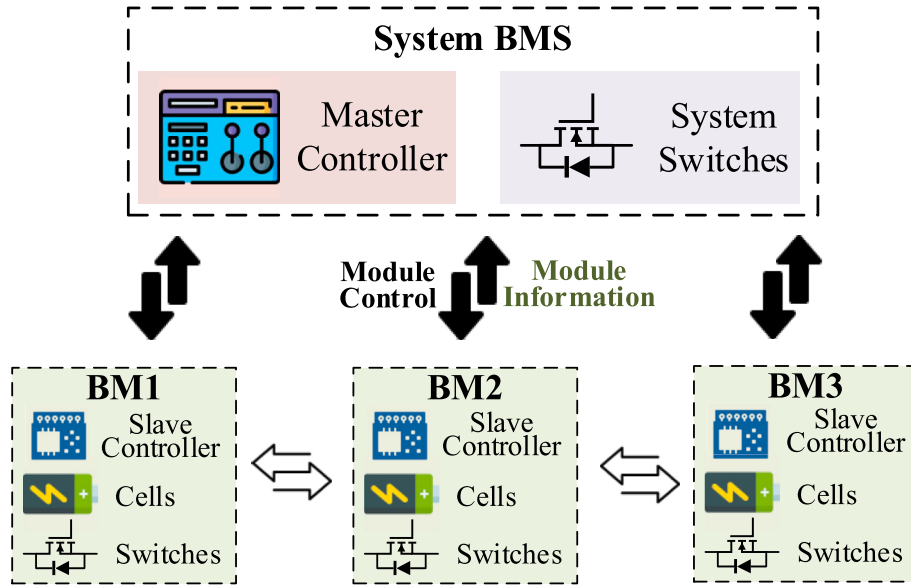


Fig. 30. Schematic diagram of distributed control architecture.

environmental interaction, eliminating the need for accurate electrochemical parameters [128,129]. Multi-agent Reinforcement Learning (MARL) frameworks are particularly effective; they decompose global optimization tasks into multiple local controllers, thereby mitigating the state-space explosion and enabling collaborative reconfiguration for large-scale RBS [130–132]. Furthermore, adaptive switching algorithms and heuristic strategies are frequently deployed at the embedded level to

balance real-time performance with implementation costs. These algorithms rely on predefined heuristic rules, such as sorting algorithms that prioritize cells based on real-time voltage or SOC. By maintaining low computational complexity, these methods achieve millisecond-level responses and ensure dynamic stability under severe load fluctuations.

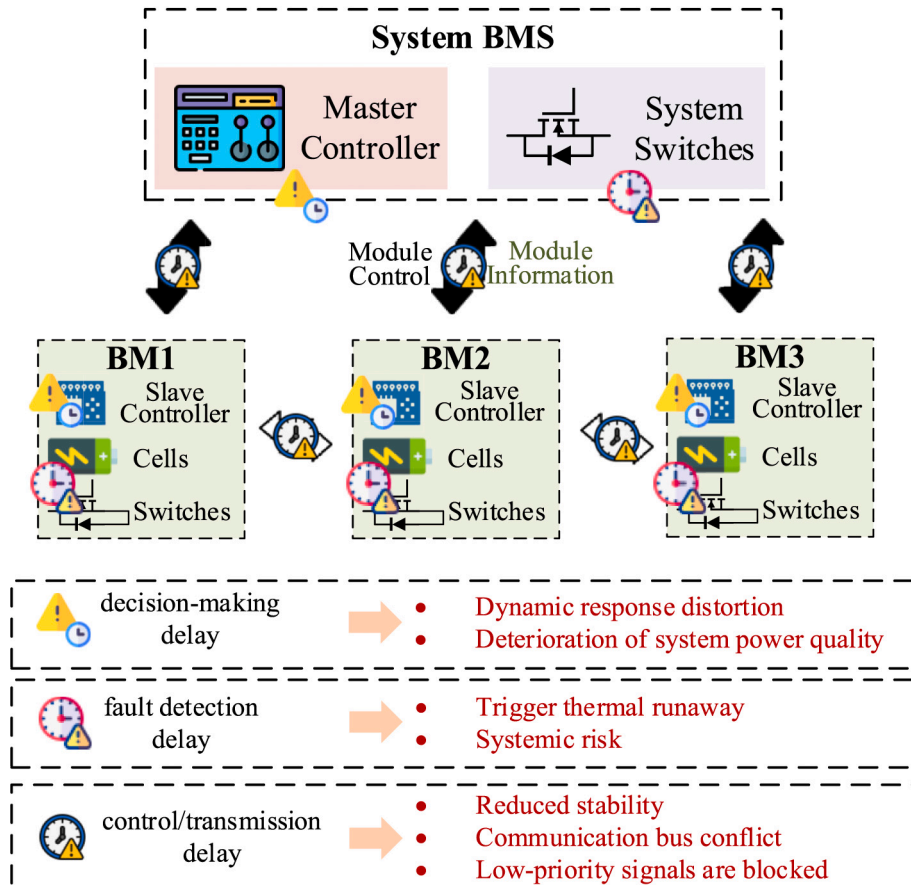


Fig. 31. Schematic diagram of signal delay impact during operation.

6.3. Impact of decision signal delay

The real-time decision-making process in RBS involves multiple stages, including data acquisition, state estimation, algorithm solving, and command execution, as shown in Fig. 31. The latency generated at each stage accumulates and propagates back to the physical layer. Decision-making delay primarily stems from the computational overhead of complex optimization algorithms on embedded processors. This lag prevents switching actions from promptly capturing the dynamic response of batteries under pulsed loads, leading to output voltage fluctuations or frequency instability. Furthermore, fault detection delay poses higher risks in RBS. Efficient isolation of internal short circuits or overheated cells in reconfiguration systems depends on rapid bypassing. Consequently, the time window from detection to execution must be minimized to prevent excessive Joule heating. If the thermal runaway threshold is exceeded, the physical layer protection will ultimately fail.

Furthermore, a significant positive correlation exists between the control signal delay τ and the number of battery nodes n . This relationship severely impacts the closed-loop stability of large-scale RBS [133]. In serial communication architectures such as CAN or CAN-FD, increasing the node count leads to a higher probability of bus contention. Due to the non-destructive arbitration mechanism, queuing delays for equalization commands or monitoring signals from low-priority nodes are significantly extended. During large-scale reconfiguration, if the transmission delay exceeds the control cycle of power converters, the system may experience limit cycle oscillations. This can further lead to severe degradation of the DC bus voltage quality [68].

7. Comparison of RBS performances

The performance and suitable application scenarios of different RBS topologies vary significantly. These differences cannot be reliably determined through direct observation alone; instead, they require systematic comparative analysis based on key performance indicators. Although numerous reconfigurable battery topologies have been proposed in the literature, they have been summarized and classified in section 2 according to the number of switches used per cell. To date, no comprehensive study has systematically compared the performance characteristics and applicable scenarios of these existing topologies. In many recent publications, topology selection appears to be made rather arbitrarily. This section provides a systematic comparative analysis of the main existing topologies in terms of cost, efficiency, reliability, and reconfiguration flexibility. It also evaluates their appropriate application scenarios and offers guidance for future research directions in reconfigurable battery systems.

7.1. Indicator analysis scenario

This study aims to investigate and compare the performance of existing RBS across different application scenarios. To achieve this, three key performance aspects are evaluated: hardware cost, efficiency, and reliability. These aspects are analyzed for representative RBS topologies in three typical energy storage applications: home energy storage, EV energy storage, and drone energy storage.

7.1.1. Home energy storage scenario (48 V, 5kWh)

In this scenario, the EVE IFR26650 battery cell is selected. N-MOSFETs are used as the series switch (S-Switch), parallel switch (P-Switch), and bypass switch (B-Switch), respectively. The key parameters of these devices are listed in Table 4. To meet the required voltage and capacity output, a minimum of 315 cells is needed. Specifically, 15 cells are connected in series to form one string, achieving a nominal voltage of 48 V ($3.2 \times 15 = 48$ V). Then, 21 such series strings are connected in parallel to obtain a total energy capacity of approximately 5.04 kWh ($5 \times 21 \times 48 = 5.04$ kWh). Alternatively, 21 cells can be connected in parallel to form one group, and 15 such groups connected in series to

Table 4

Parameters of components in home energy storage scenario.

Component	Description	Value	Price
Battery Cells	EVE IFR26650	3.2 V 5 Ah	\$2.4
	Internal Resistance	12mΩ	
S-Switch MOSFET	ST STB120N4F6	$V_{DS} = 40$ V $I_D = 80$ A $R_{DS} = 4$ mΩ	\$2.1
P-Switch MOSFET	ST STH170N8F7-2	$V_{DS} = 80$ V $I_D = 120$ A $R_{DS} = 2.8$ mΩ	\$2.6
B-Switch MOSFET	ST STH170N8F7-2	$V_{DS} = 80$ V $I_D = 120$ A $R_{DS} = 2.8$ mΩ	\$2.6

achieve the same voltage and capacity. In a double-layer topology, the outer and inner layers experience different voltage and current stresses. Therefore, the outer-layer switches (O-Switch) must have higher voltage and current ratings than the inner-layer switches. The specifications selected for the outer-layer switch are provided in Table 5. The hardware cost of various existing reconfigurable battery topologies to achieve a 48 V, 5 kWh system in a home energy storage scenario is shown in Fig. 29a.

7.1.2. Electric vehicle (EV) energy storage scenario (400 V, 50kWh)

In this scenario, the FLYOUNG IFR32700 battery is selected as the cell unit. It has a rated voltage of 3.2 V and a rated capacity of 6.5 Ah. N-MOSFETs are chosen as the switching devices. The relevant parameters of the battery cells and switches are listed in Table 6. To meet the energy storage requirements of the electric vehicle, at least 2500 battery cells are required. Specifically, 125 cells are connected in series to achieve a nominal voltage of 400 V ($3.2 \times 125 = 400$ V). Then, 20 such series strings are connected in parallel to obtain a total capacity exceeding 50 kWh ($400 \times 6.5 \times 20 = 52$ kWh). In the double-layer topology applied to this scenario, the outer-layer switches must withstand higher voltage and current stresses. The selected parameters for the outer-layer switches are provided in Table 7. The hardware cost of various existing reconfigurable battery topologies to achieve a 400 V, 50 kWh system for EV energy storage is illustrated in Fig. 29b.

7.1.3. Drone energy storage scenario (22.2 V, 222 Wh)

Samsung INR21700 50E batteries are used as the battery cells, each with a rated voltage of 3.7 V and a rated capacity of 5 Ah, N-MOSFETs are employed. The relevant parameters of these devices are shown in Table 8. To meet the energy storage requirements of this drone, at least 12 batteries are needed, with 6 battery cells connected in series to achieve the required voltage output ($3.7 \times 6 = 22.2$ V), and then such series strings connected in parallel to meet the total capacity requirement ($5 \times 2 \times 22.2 = 222$ Wh). In the double-layer topology application of this scenario, the selection parameters of the outer-layer switch are shown in Table 8. The cost required for the existing reconfigurable battery topology in a 22.2 V, 222 Wh drone scenario is shown in Fig. 29 (c).

In this drone scenario, the Samsung INR21700 50E battery is selected as the cell unit. Each cell has a rated voltage of 3.7 V and a rated capacity of 5 Ah. N-MOSFETs are employed as the switching devices. The key parameters of the battery cells and switches are listed in Table 7. To meet the energy storage requirements of the drone, a minimum of 12 battery cells is required. Six cells are connected in series to achieve the target voltage of 22.2 V ($3.7 \times 6 = 22.2$ V). Two such series strings are

Table 5

Parameters of the O-switch in home energy storage scenario.

Component	Description	Value	Price
O-Switch MOSFET	ST STH310N10F7-2	$V_{DS} = 100$ V $I_{DS} = 180$ A $R_{DS} = 2.3$ mΩ	\$2.6

Table 6

Parameters of components in EV energy storage scenario.

Component	Description	Value	Price
Battery Cells	FLY IFR32700	3.2 V 6.5 Ah	\$3.2
	Internal Resistance	10mΩ	
S-Switch MOSFET	Infineon IRFS7440	$V_{DS} = 40 \text{ V}$ $I_D = 120 \text{ A}$ $R_{DS} = 2.5\text{m}\Omega$	\$2.1
P-Switch MOSFET	Infineon IPB020N08N5	$V_{DS} = 80 \text{ V}$ $I_D = 173 \text{ A}$ $R_{DS} = 2\text{m}\Omega$	\$2.8
B-Switch MOSFET	Infineon IPB020N08N5	$V_{DS} = 40 \text{ V}$ $I_D = 120 \text{ A}$ $R_{DS} = 2.5\text{m}\Omega$	\$2.8

Table 7

Parameters of the O-switch in EV energy storage scenario.

Component	Description	Value	Price
O-Switch MOSFET	Infineon IPB039N10N3	$V_{DS} = 100 \text{ V}$ $I_{DS} = 160 \text{ A}$ $R_{DS} = 3.9\text{m}\Omega$	\$2.8

Table 8

Parameters of components in drone energy storage scenario.

Component	Description	Value	Price
Battery Cells	Samsung INR21700	3.7 V 5 Ah	\$2.8
	Internal Resistance	35mΩ	
S-Switch MOSFET	AOS AOD4186	$V_{DS} = 40 \text{ V}$ $I_{DS} = 35 \text{ A}$ $R_{DS} = 15\text{m}\Omega$	\$2
P-Switch MOSFET	AOS AON6230	$V_{DS} = 40 \text{ V}$ $I_{DS} = 85 \text{ A}$ $R_{DS} = 1.44\text{m}\Omega$	\$2.4
B-Switch MOSFET	AOS AON6230	$V_{DS} = 40 \text{ V}$ $I_{DS} = 85 \text{ A}$ $R_{DS} = 1.44\text{m}\Omega$	\$2.4

then connected in parallel to provide the total energy capacity of 222 Wh ($5 \times 2 \times 22.2 = 222 \text{ Wh}$). In the double-layer topology applied to this scenario, the outer-layer switches experience higher voltage and current stresses. The selected parameters for the outer-layer switches are given in Table 9. The hardware cost of various existing reconfigurable battery topologies to achieve a 22.2 V, 222 Wh system for drone energy storage is presented in Fig. 29c.

7.2. Cost analysis

Relative to a reference system, the cost estimates of all compared configurations are expressed based on the main system components [134]. The primary components of a RBS are cells, switches, and sensors. Therefore, cost estimation for both RBS and traditional battery systems focuses primarily on these three elements: cells, switches, and sensors. However, cost differences among RBS topologies in the same application scenario arise mainly from variations in the number of switches used. Consequently, the cost analysis of existing RBS topologies concentrates primarily on the cost of switching devices.

Due to the existence of various evolutionary variants of three-switch and four-switch topologies, a systematic cost evaluation is necessary. This study first categorizes the existing three-switch and four-switch

Table 9

Parameters of the O-switch in drone energy storage scenario.

Component	Description	Value	Price
O-Switch MOSFET	AOS AONS66612	$V_{DS} = 60 \text{ V}$ $I_{DS} = 100 \text{ A}$ $R_{DS} = 3.8\text{m}\Omega$	\$3.9

topologies and analyzes their costs accordingly. Within each category, the topology offering the best cost-performance ratio is selected. Finally, a comprehensive cost comparison of different switch-based topologies is performed. It should be noted that the category-based cost analysis of three-switch and four-switch topologies focuses exclusively on the EV application scenario. This focus enables a targeted exploration of topologies with the highest cost-effectiveness.

Due to the significant price volatility, a stable and reliable cost analysis of reconfigurable topologies is challenging. Therefore, the system cost is defined as follows:

$$Cost_{\text{total}} = C_{\text{cell}}(n + m \cdot \alpha) \quad (16)$$

where C_{cell} represents the price of a battery cell, n and m denote the quantities of batteries and switches used, respectively. α represents the cost ratio between the switch and the battery, defined as $\alpha = C_{\text{switch}}/C_{\text{cell}}$. To clearly illustrate the cost implications of each topology, this analysis estimates the costs of the various topologies using market prices from 2025 to 2026.

7.2.1. Cost analysis of three-switch topology

Six variants of three-switch topologies have been proposed to date, as shown in Figs. 6–9. The costs of existing three-switch topologies in EV energy storage applications are presented in Fig. 32. This cost analysis specifically considers the use of single MOSFETs and bidirectional MOSFETs, which explains the significant cost differences observed among the various three-switch topologies in the figure. According to Fig. 32, the three-switch RBS topologies that support the largest number of configuration options while maintaining excellent cost performance correspond to the design shown in Fig. 9. Therefore, subsequent evaluations of key performance indicators for existing three-switch topologies will primarily focus on the topology presented in Fig. 9.

7.2.2. Cost analysis of four-switch topology

Six evolved four-switch RBS topologies have been developed, as shown in Figs. 13–18. Since the topology in Fig. 18 is intended primarily for AC battery systems, it is not included in the present cost analysis of four-switch topologies. Cost comparisons for the four-switch topologies in EV energy storage applications are given in Fig. 33.

As shown in Fig. 33, the existing four-switch topologies generally exhibit comparable cost performance. The topology illustrated in Fig. 17 is a notable exception, as it incurs significantly higher costs due to the use of multiple bidirectional MOSFETs. Meanwhile, the four-switch topology presented in Fig. 13 is derived from the three-switch topology

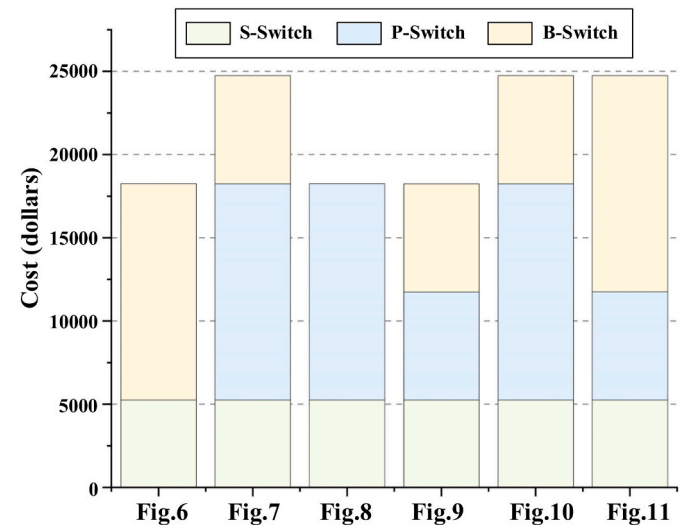


Fig. 32. Comparison chart of three switches topology cost in the EV energy storage scenarios.

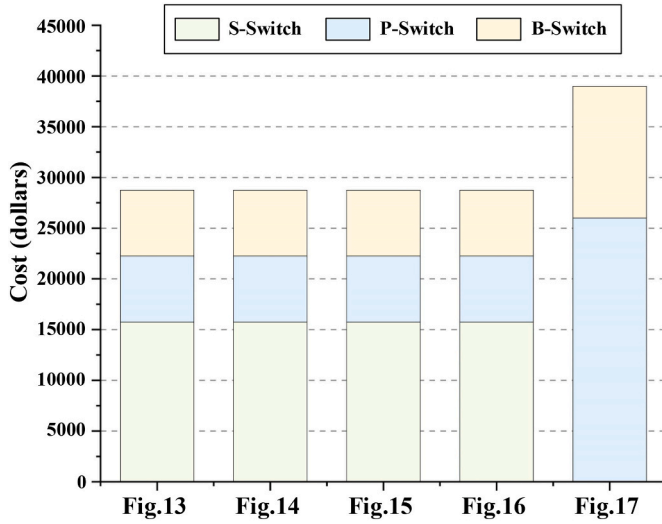


Fig. 33. Comparison chart of four switches topology cost in the EV energy storage scenario.

shown in Fig. 9. To better investigate the application scenarios and performance of different switch-type topologies, subsequent evaluations of key performance indicators for existing four-switch topologies will primarily focus on the design depicted in Fig. 13.

7.2.3. Cost analysis of the existing topology

The existing RBS topologies are summarized and introduced in Section 2. In the following sections, a cost estimation and detailed analysis of these topologies are presented for three energy storage application scenarios: home energy storage, EV energy storage, and drone energy storage.

As shown in Fig. 34, the cost of constructing RBS is higher than that of traditional battery systems. Traditional battery systems feature a fixed architecture without switching devices for topology reconfiguration. This absence of switches also results in lower fault tolerance. Furthermore, traditional systems typically monitor only the voltage and current of each series branch, whereas RBS requires precise monitoring of voltage and current at the individual cell level [135].

Compared with home energy storage and EV applications, the cost increase associated with each topology in drone energy storage scenarios is relatively modest (see Fig. 34c). This suggests that drone applications require fewer switches to achieve the necessary system complexity. In contrast, EV energy storage applications exhibit higher cost levels across topologies (see Fig. 34b), driven by the need for high-power control. Higher power demands impose stricter requirements on energy management. Consequently, in high-power scenarios such as EVs, the selected battery system topology should provide enhanced control and adjustment capabilities. In medium to large-scale energy storage applications (see Fig. 34a and b), the cost difference between double-layer and single-layer topologies using the same switch type is generally small. Nevertheless, due to their structural characteristics, double-layer topologies may offer greater overall advantages in medium to large-scale applications.

7.2.4. Cost performance of various topologies under price fluctuations

The cost analyses above rely on fixed market prices. Variations in these prices can influence the outcomes. To assess the sensitivity of the total system cost to price fluctuations, this paper evaluates the cost using the model given in Eq. 16. The study specifically examines how the total cost changes with the switch-to-battery price ratio, as shown in Fig. 35.

As shown in Fig. 35, the cost of more complex topologies increases more significantly with rising switch-to-battery price ratios α , regardless of energy storage scale. In other words, when switch prices approach or

exceed battery prices, the cost differences among topologies become more pronounced. This observation indicates that, despite potential fluctuations in device prices, the relative cost performance of each topology remains consistent with the findings obtained using fixed reference prices. This inherent structural characteristic is independent of currency value variations.

However, as the proportion α approaches zero, the future price of switches becomes negligible compared to the battery price. In this case, topology complexity no longer represents a major cost factor. Consequently, the application costs of different topologies tend to converge. As a result, the flexibility advantages offered by more complex topologies become increasingly evident.

7.3. Efficiency analysis

Efficiency is a key performance indicator for RBS. It directly reflects the system's ability to utilize energy effectively [136]. Compared with traditional fixed series-parallel battery systems, RBS can achieve dynamic optimization of energy transfer through topological reconfiguration. However, the switching devices introduce switching losses. These losses reduce the overall system efficiency. Therefore, a thorough efficiency analysis of an RBS must include a detailed evaluation of its switching losses.

The formula for calculating the conduction loss of a MOSFET is:

$$P_{\text{mos}}^{\text{loss}} = I_{\text{DS}}^2 R_{\text{DS}}^{\text{ON}} \quad (17)$$

where I_{DS} represents the current flowing between the drain and source of the MOSFET, that is, the current through the device. $R_{\text{DS}}^{\text{ON}}$ denotes the on-resistance of the MOSFET. Its value is obtained from the datasheet of the selected MOSFET model. Due to its internal structure, every MOSFET includes a body diode. During circuit operation, current may flow through this body diode. This conduction path must also be considered in the analysis. In this case, the conduction loss of the body diode is:

$$P_{\text{Bdiode}}^{\text{loss}} = I_{\text{DS}} V_{\text{SD}} \quad (18)$$

where V_{SD} is the forward voltage of the body diode. Therefore, the operational loss of the MOSFET can be defined as the sum of its own conduction loss $P_{\text{mos}}^{\text{loss}}$ and the conduction loss of the body diode $P_{\text{Bdiode}}^{\text{loss}}$:

$$P_{\text{mos}} = \sum P_{\text{mos}}^{\text{loss}} + \sum P_{\text{Bdiode}}^{\text{loss}} \quad (19)$$

In practical operation, the efficiency of RBS is influenced by multiple factors. Besides the operational losses introduced by the MOSFETs, the conduction losses caused by the internal resistance of the batteries R_{in} also have a substantial impact on overall system efficiency. The conduction loss of the battery is expressed as:

$$P_{\text{cell}}^{\text{loss}} = I_{\text{B}}^2 R_{\text{in}} \quad (20)$$

where I_{B} represents the battery current, with a positive value indicating battery discharge and a negative value indicating battery charging. The efficiency estimation analysis of the RBS in this study primarily focuses on the system discharge scenario. Therefore, the battery current I_{B} should be positive. The system loss P_{loss} is determined jointly by the battery conduction loss $P_{\text{cell}}^{\text{loss}}$ and the MOSFET operating loss P_{mos} :

$$P_{\text{loss}} = N_{\text{cell}} P_{\text{cell}}^{\text{loss}} + P_{\text{mos}} \quad (21)$$

where N_{cell} represents the number of batteries participating in operation. At this point, the output power of the system batteries is defined as: $P_{\text{cell}} = N_{\text{cell}} \bullet I_{\text{B}} \bullet V_{\text{cell}}$. Based on the operational losses of the system and the output power of the system batteries, the operational efficiency of the RBS is expressed as:

$$\eta = \frac{P_{\text{cell}}}{P_{\text{cell}} + P_{\text{loss}}} \quad (22)$$

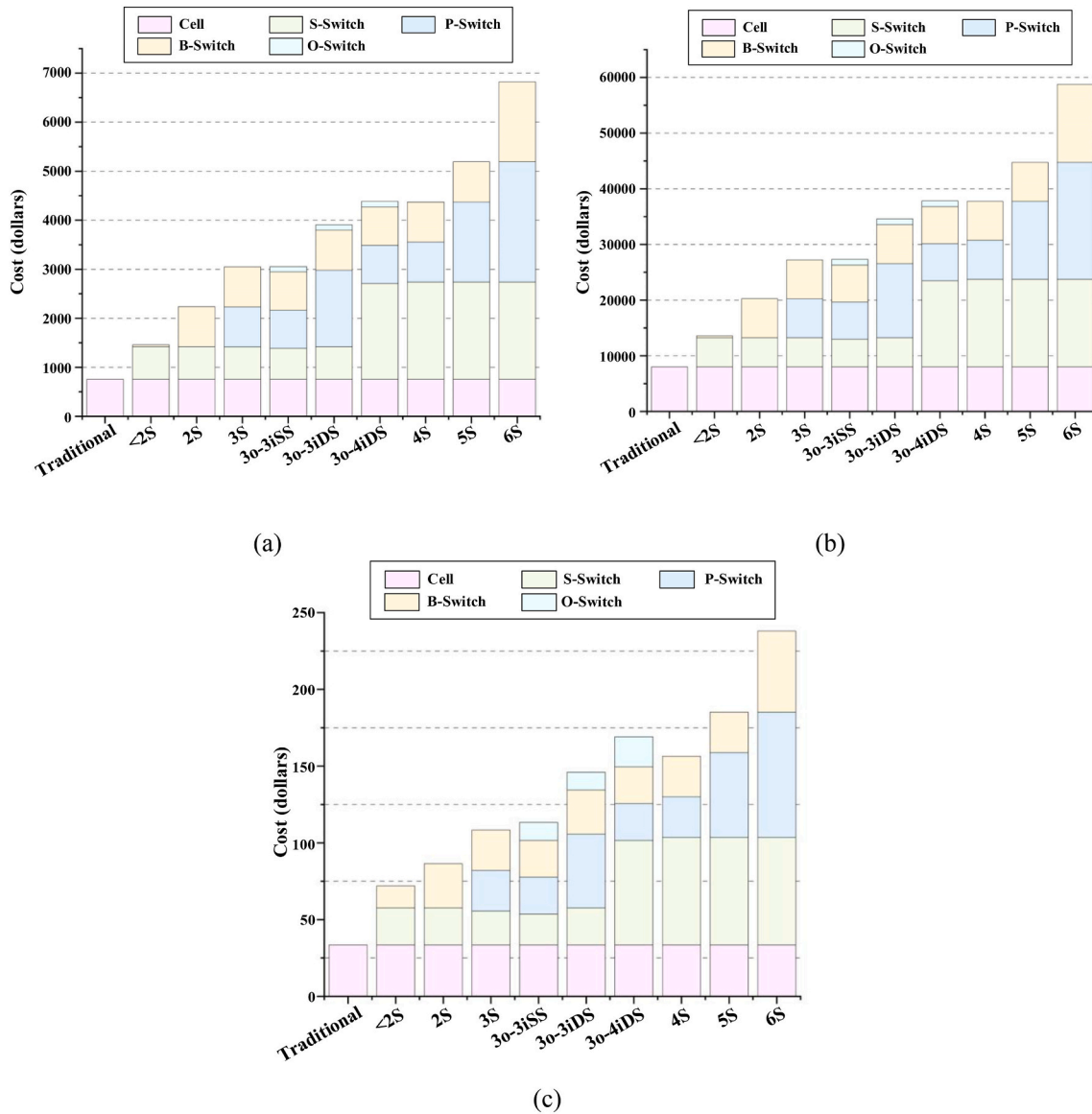


Fig. 34. Comparison chart of reconfigurable battery system cost: (a) Home energy storage scenario. (b) EV energy storage scenario. (c) Drone energy storage scenario.

Fig. 36 shows the efficiency variations of existing RBS topologies across different application scenarios and numbers of batteries. The relevant calculation parameters are given in Tables 4 to 9. As the number of batteries increases, or equivalently, as the energy storage scale grows, the overall output efficiency of the RBS generally improves. This improvement occurs because the system's output power depends on the combined output power of the batteries. A larger number of batteries delivers higher total output power. Although the associated losses also increase, this rise remains negligible compared to the gain in overall system output power.

As clearly shown in Fig. 36, the five-switch topology achieves high output efficiency comparable to that of low-switch topologies, such as the Self-X and two-switch configurations. At the same time, reducing the number of batteries leads to a significant decrease in output efficiency for most reconfigurable topologies. In medium to large-scale energy storage applications, however, the dual-layer topology remains largely unaffected by the number of batteries. It maintains stable and highly efficient output performance (see Fig. 36a and b). In contrast, in small-scale energy storage applications, the output efficiency of the dual-layer topology is noticeably sensitive to the number of batteries. Meanwhile,

the efficiency degradation of the two-switch topology becomes much less pronounced (see Fig. 36c). In this small-scale scenario, the choice of topology has a relatively minor influence on overall output efficiency. Consequently, selecting a simpler topology is often more suitable to meet the application requirements.

The analysis of Fig. 36 can be summarized as follows:

- 1) Although the increase in switches will also lead to a corresponding rise in system losses, this does not mean that topologies with more switches are inherently less efficient. A five-switch topology can achieve high-efficiency output comparable to that of a Self-X topology.
- 2) In medium to large-scale energy storage scenarios, double-layer topologies exhibit significantly superior stable and efficient output performance compared to single-layer topologies using the same switch type. This enables the system to maintain high output efficiency even when the number of batteries is reduced due to faults or other factors.

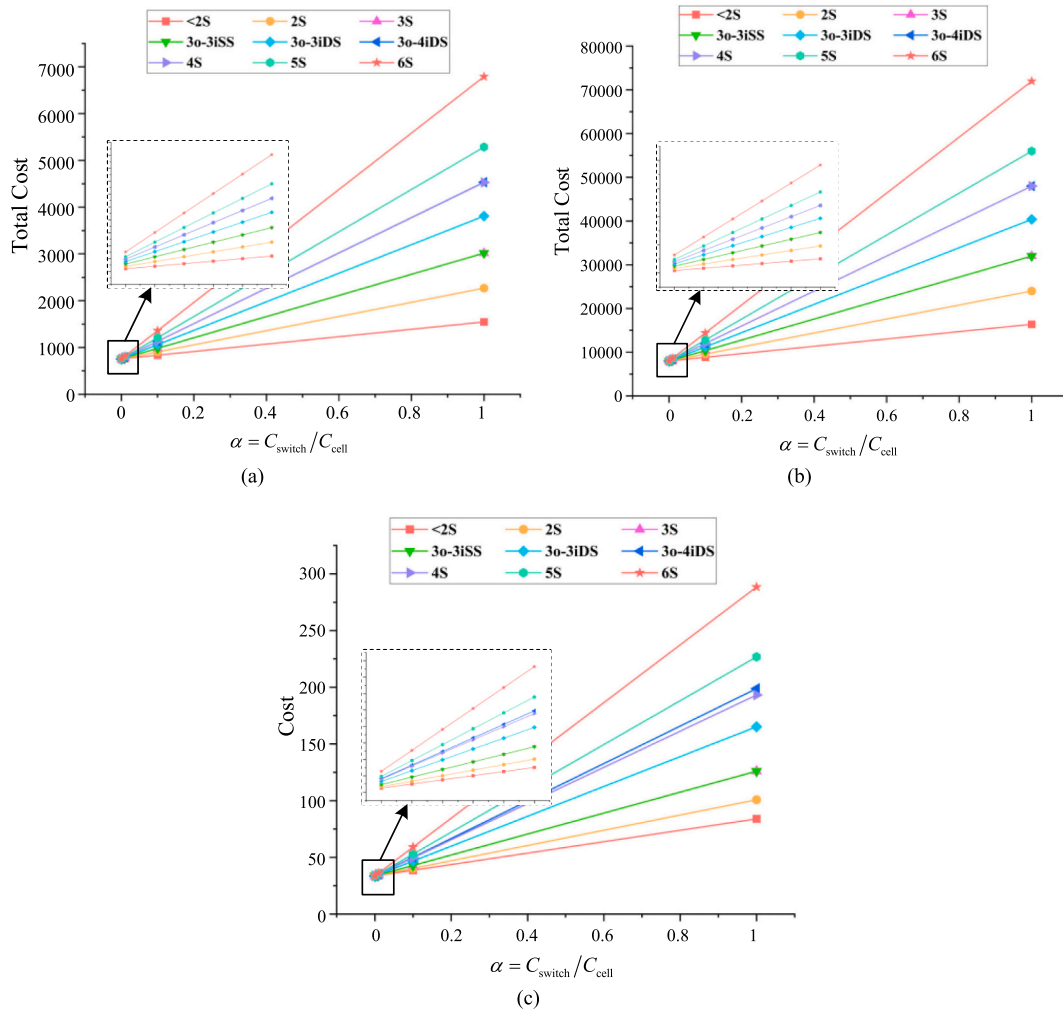


Fig. 35. Comparison chart of reconfigurable battery system costs under component price changes: (a) Home energy storage scenario; (b) EV energy storage scenario; (c) Drone energy storage scenario.

3) Simpler topologies are more suitable for low-scale energy storage scenarios, such as the drone energy storage scenario selected in this study.

7.4. Reliability analysis

In traditional battery systems, the fixed series connection of battery cells leads to the “weakest link effect”. This phenomenon significantly reduces overall system reliability, particularly when a single cell fails. In such cases, conventional fixed battery packs struggle to maintain efficient operation [137]. In contrast, reconfigurable battery systems offer dynamic redundancy through topology adjustment. They can effectively bypass faulty cells and sustain stable operation. As a result, these systems achieve substantially higher reliability.

The reliability of a system is generally defined as the probability that the system will perform its intended function within the specified conditions for a given period of time [138,139]. Based on the assumption that component failure rates are independent of time [140], the reliability formula for a device over time t is expressed as:

$$R(t) = e^{-\int_0^t \lambda(\tau) d\tau} = e^{-\lambda t} \quad (23)$$

where λ represents the failure rate of the system components. The main components of a reconfigurable battery system are primarily the battery cells and the switching devices.

In practical operation, the battery and the switch can be considered

as a series configuration. Therefore, the reliability of the reconfigurable battery system can be mainly expressed as:

$$R_S(t) = R_B(t) \times R_M(t) \quad (24)$$

where $R_S(t)$, $R_B(t)$ and $R_M(t)$ represent the reliability of the reconfigurable battery system, all batteries, and the switches, respectively.

Similarly, the operational failure rate of the system primarily refers to the failure rates of all battery cells and MOSFETs within the system:

$$\lambda_S = \lambda_B + \lambda_M \quad (25)$$

where λ_S , λ_B and λ_M represent the failure rates of the reconfigurable battery system, battery cells, and MOSFETs, respectively.

The failure rates of the components are estimated using the MIL-HDBK-217F (N2) standard, a model that is internationally recognized for reliability calculations [141–143]. According to this method, the formula for calculating the failure rate of system components is as follows:

$$\lambda = \lambda_b \pi_E \pi_Q \prod_i \pi_i \quad (26)$$

where λ_b is the base failure rate, π_E is the environmental factor, π_Q is the limiting factor, and π_i is the component characteristic. Specifically, based on this reliability algorithm, the failure rates of the battery cell and the MOSFET switch are represented by eqs. 26 and 27, respectively.

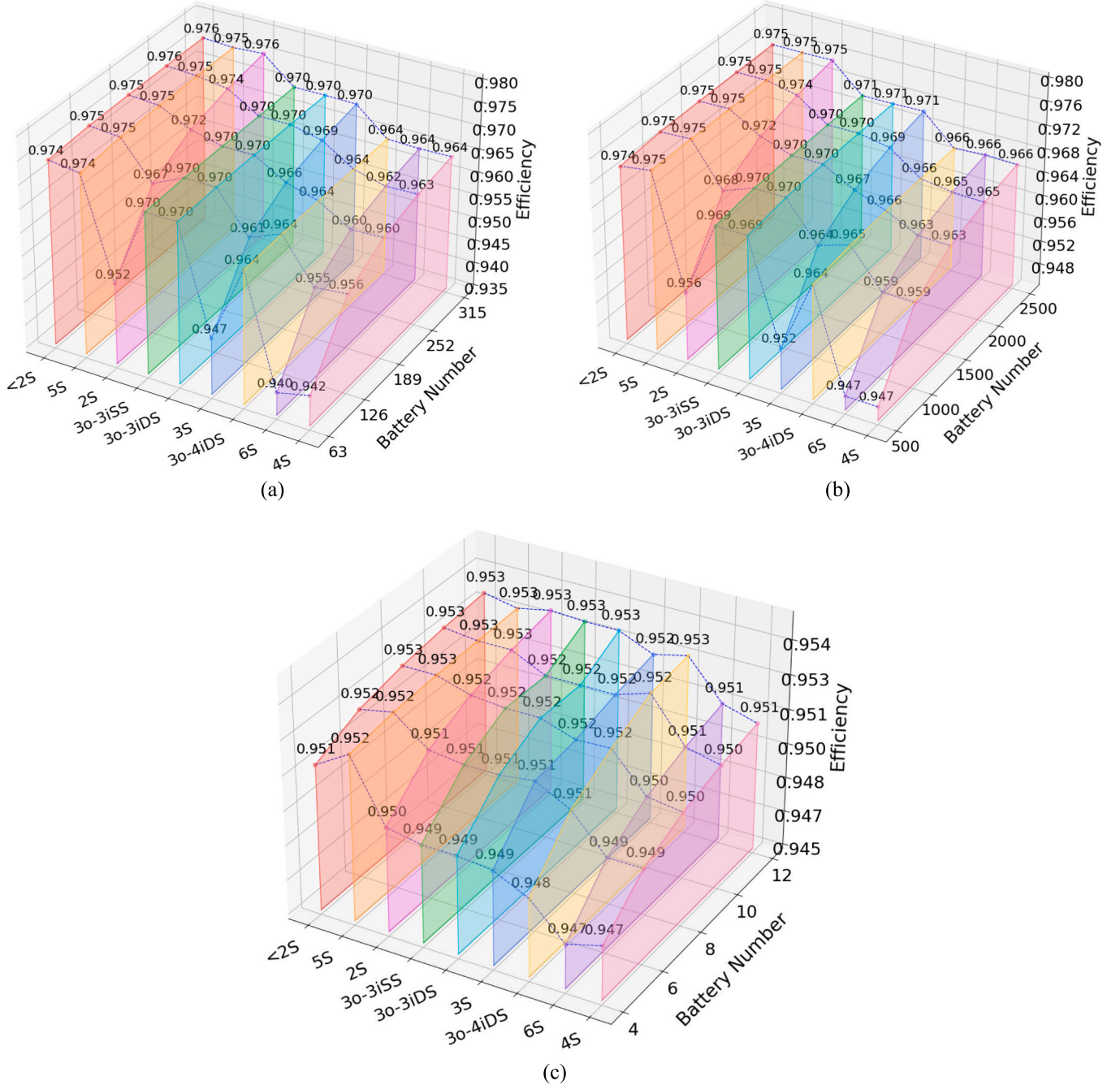


Fig. 36. System efficiency with existing battery topologies with different battery counts: (a) Home energy storage scenario. (b) EV energy storage scenario. (c) Drone energy storage scenario.

$$\lambda_B = \lambda_b \pi_E \pi_Q \quad (27)$$

$$\lambda_M = \lambda_b \pi_E \pi_Q \pi_A \pi_T \quad (28)$$

where π_A and π_T are the application factor and the temperature factor, respectively. The temperature factor mainly depends on the temperature T_j of the MOSFET and is a key factor affecting the failure rate of the MOSFET [143,144]. The formula for calculating π_T is:

$$\pi_T = \exp\left(-1925 \cdot \left(\frac{1}{T_j + 273}\right) - \frac{1}{298}\right) \quad (29)$$

When multiple devices or modules of the same type are present, each unit has an identical and independent failure rate. Therefore, these devices or modules can be modeled as a k-out-of-n system for reliability

calculation [145]. In this case, the reliability of the system is calculated using the following formula:

$$R_M(t) = \sum_{i=k}^n \binom{n}{i} R_{\text{single}}^i(t) [1 - R_{\text{single}}(t)]^{n-i} \quad (30)$$

where $R_{\text{single}}(t)$ represents the reliability of a single device or module, n denotes the total number of devices or modules in the system, k is the threshold number of devices or modules required to maintain normal system operation, and i is the number of devices or modules functioning properly, satisfying $k \leq i \leq n$. Table 10 provides a detailed description of the parameters used for failure rate calculations [146,147].

Compared with traditional battery systems, a major advantage of RBS is their ability to maintain efficient operation under derated con-

Table 10
Failure rate calculation parameters.

Parameter	Value
Temperature	55 °C
λ_b -battery	0.4×10^{-6}
λ_b -MOSFET	0.012×10^{-6}
π_E	1.0
π_Q	8.0
π_A	8.0

ditions. These conditions may arise from battery failures or significant deviations in cell states. Therefore, reliability analysis of an RBS should account for derated operation capability. This factor plays a critical role in ensuring stable system performance. At this time, the system's derated operation capability R_{RO} can be expressed as:

$$R_{RO} = \alpha_1 \frac{Q_{RE}}{Q_{TO}} + \alpha_2 \frac{V_{RE}}{V_{TO}} \quad (31)$$

where α_1 and α_2 are weighting coefficients, with $\alpha_1 + \alpha_2 = 1$. Q_{RE} is the output capacity of the derated system, Q_{TO} is the overall system

capacity, V_{RE} is the output voltage of the derated system, and V_{TO} is the overall system voltage.

Therefore, the reliability of the RBS should comprehensively consider the system's reduced operational capability R_{RO} and the component failure rate λ_{sys} .

$$R_{sys} = f(\lambda_{sys}, R_{RO}) = \omega_1 e^{-\lambda_{sys} t} + \omega_2 R_{RO} \quad (32)$$

where ω_1 and ω_2 are the corresponding weights, indicating the significance of this parameter in reliability assessment. R_{sys} represents the system reliability.

Fig. 37 presents the reliability of existing topologies for different numbers of batteries across various application scenarios. The results in Fig. 37 indicate that topological reliability does not always increase with the number of switches. As shown in Fig. 37a and b, the five-switch topology consistently achieves the highest reliability regardless of battery count. In contrast, the four-switch topology exhibits lower reliability than both the three-switch topology and the double-layer topology.

Additionally, in medium to large-scale energy storage applications, the double-layer topology exhibits superior reliability compared with

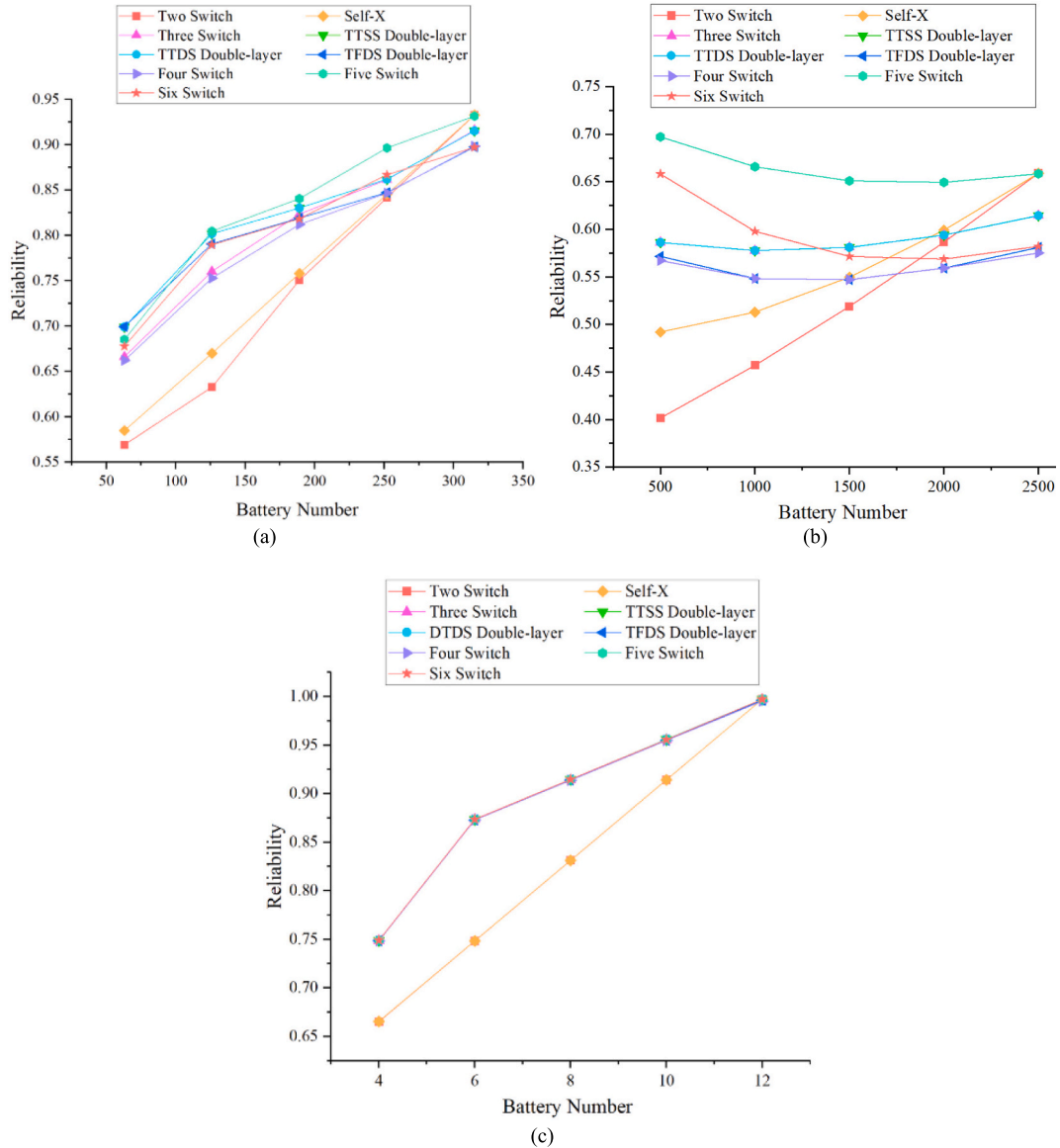


Fig. 37. System reliability with existing battery topologies with different battery counts: (a) Home energy storage scenario. (b) EV energy storage scenario. (c) Drone energy storage scenario.

single-layer topologies of the same type. The Self-X topology and the two-switch configuration have simple structures and limited ability to handle system derating. As a result, their reliability varies noticeably across the three scenarios. In large-scale energy storage applications, multi-switch topologies become more complex. The number of batteries and switching devices increases significantly. This leads to a higher overall system failure rate. Consequently, system reliability decreases when the number of batteries ranges from 500 to 1500 (see Fig. 37b). However, multi-switch topologies offer strong derating capability. When the number of batteries exceeds 1500 and reaches 2500, system reliability shows a gradual upward trend (see Fig. 37b). In small-scale energy storage scenarios, fewer batteries are used. The reliability difference between multi-switch and low-switch topologies mainly stems from topology complexity. Reconfiguration capability is not a dominant factor. Therefore, the reliability of multi-switch and low-switch topologies is essentially similar in this scenario (see Fig. 37c). Based on the analysis of Fig. 37, the following summary can be made:

- 1) Topological reliability does not increase solely with the number of switches. More switches do not always result in higher reliability. In the analysis of the three application scenarios, the five-switch topology exhibits the highest reliability.
- 2) In medium to large-scale energy storage scenarios, the dual-layer topology demonstrates higher reliability than similar single-layer topologies. Adjusting a single-layer topology into a double-layer form can effectively enhance system reliability.
- 3) In small-scale energy storage scenarios, the reconfiguration capability of the topology is not a decisive factor. Therefore, in such applications, selecting a simple topology can meet basic reliability requirements and ensuring low costs.

7.5. Flexibility analysis

The flexibility of an RBS topology reflects the diversity of its possible reconfiguration methods. A highly flexible topology supports a larger number of ways to alter the battery connection states. In general, system flexibility is directly linked to the range of series-parallel configurations that the battery pack can achieve. Existing studies have not yet provided a quantitative measure of flexibility for RBS topologies. In this paper, each RBS topology is modeled as a directed graph. Using a small number of battery cells as an example, each valid path from the positive terminal to the negative terminal through the battery cells represents a feasible reconfiguration state supported by the topology, as illustrated in Fig. 38. To enable a systematic analysis of flexibility in existing RBS topologies, we quantify topology flexibility as the total number of valid paths in the corresponding directed graph that connect the positive and negative terminals.

In the directed graph model, each node corresponds to a battery cell. Edges between nodes represent switch connections. The positive and negative terminals are modeled as the source and sink nodes of the graph. Through computational analysis of the directed graph, the total number of valid paths from the positive terminal to the negative terminal is determined for each topology. These results are summarized in Table 11.

According to the data in Table 11, multi-switch topologies exhibit greater flexibility than low-switch topologies. Among single-layer configurations, the five-switch topology shows significantly higher flexibility than the others. Furthermore, double-layer topologies achieve higher overall flexibility compared to single-layer topologies. In addition, different combinations of three-switch topologies can substantially increase the flexibility of the system.

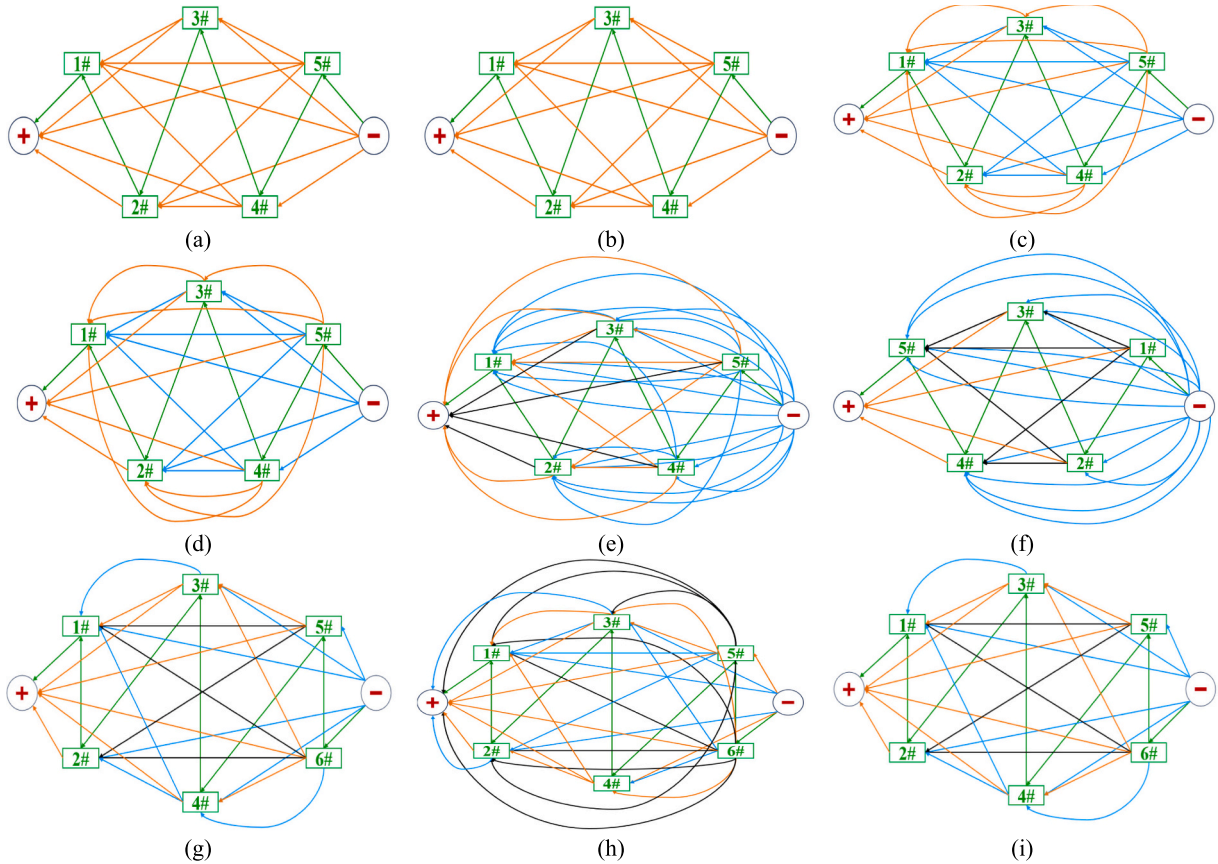


Fig. 38. The directed graph corresponding to the current topology: (a) Less than Two Switches Topology. (b) Two Switches Topology. (c) Three Switches Topology. (d) Four Switches Topology. (e) Five Switches Topology. (f) Six Switches Topology. (g) TTSS Double-layer Topology. (h) TTDS Double-layer Topology. (i) TFDS Double-layer Topology.

Table 11

The total number of valid paths in the existing topology.

Topology	Number of Switches	Number of paths
Self-X	10	31
Two-Switch	10	31
Three-Switch	12	49
Four-Switch	17	49
Five-Switch	22	113
Six-Switch	32	57
TTSS Double-layer	15	80
TTDS Double-layer	19	143
TFDS Double-layer	21	80

7.6. Structural thermal impact analysis

Clearly, in RBS, the high-density structure enabled by multi-switch topologies significantly enhances the flexibility of circuit reconfiguration and scheduling. However, the incorporation of switching devices also introduces complex thermal coupling mechanisms. Unlike traditional battery systems, where heat generation is primarily driven by electrochemical reactions within individual cells, high-density reconfigurable topologies incorporate power semiconductor devices such as MOSFETs. These devices produce significant switching losses and conduction losses during high-frequency operation and large-current conduction [81]. The switching devices are tightly integrated and positioned in close proximity to the battery cells. As a result, localized point heat sources from the MOSFETs combine with the distributed volumetric heat sources of the battery cells. This interaction produces a composite thermal field [148,149]. This thermal superposition effect not only raises the overall baseline temperature of the system but also causes pronounced local hot spots at the cell terminals and interconnection regions. Consequently, it seriously compromises the reliability of the power electronics and the thermal stability of the battery cells [5].

Furthermore, as topological density increases, the physical design space available for thermal management becomes severely constrained. For example, in a four-switch topology configuration, the highly dense gate drive circuits and sensor arrays occupy space that would otherwise be used for cooling channels. In forced-air or liquid cooling schemes, the complex wiring and densely packed PCB components significantly increase the flow resistance of the coolant. This makes it difficult for the cooling medium to reach all hot spots uniformly. As a result, localized regions with inadequate cooling, known as thermal blind zones, are formed [150]. Therefore, to achieve higher power density, the packaging design of high-density topologies must strike a careful balance between electrical insulation and efficient heat conduction. For high-density RBS in particular, the design must simultaneously satisfy functional requirements and compactness goals while ensuring well-planned and effective heat dissipation paths.

7.7. Summary

The comparison of cost, efficiency, reliability, and flexibility for existing topologies is summarized in Fig. 39, based on the preceding analysis. As shown in the figure, multi-switch topologies generally provide better overall system performance. However, this improvement comes at the expense of higher costs. The trend is especially evident in five-switch topologies and dual-layer architectures. Moreover, overall topology performance is not determined solely by the degree of complexity. Instead, it is closely linked to the types of reconfiguration supported and the number of available topological paths.

In medium to large-scale energy storage applications, the five-switch topology exhibits superior efficiency and reliability compared to the dual-layer topology. However, its cost is substantially higher. The double-layer topology, by contrast, achieves performance levels comparable to those of the five-switch topology while maintaining significantly lower costs. Consequently, the dual-layer topology demonstrates

better overall performance than single-layer topologies in energy storage applications (see Fig. 39a and b). This finding indicates that adopting a dual-layer configuration can effectively improve the comprehensive performance of a single-layer topology.

In small-scale energy storage applications, the requirements for reconfiguration capability of the RBS are relatively low. Therefore, the selection of RBS architecture should prioritize cost and efficiency. Compared with other options, simple topologies that offer low cost and high efficiency, such as the Self-X and two-switch topologies, are more suitable for scenarios with modest energy storage demands.

7.8. Discussion on future development directions

This analysis shows that double-layer topologies provide superior performance in high-capacity energy storage applications due to their structural decoupling. By separating the switching network into intra-module and inter-module levels, this architecture effectively resolves the trade-off between the number of switches and the available control degrees of freedom in large-scale systems. As the demand for large-scale RBS continues to grow, double-layer configurations are expected to emerge as a major research direction in the future. Future work should investigate the dynamic adjustment of topological granularity according to specific application requirements. For example, simpler topologies can be implemented at the cell level to reduce conduction losses. At the same time, the integration of high-frequency power electronic switches enables real-time series-parallel reconfiguration and fault isolation at the cell level. This strategy eliminates the weakest link effect and maximizes the capacity utilization of individual cells. At the system level, highly flexible reconfigurable topologies combined with modular clustering techniques can be employed to achieve precise control of power flow.

8. Conclusions

Compared with traditional battery systems that employ fixed configurations, reconfigurable topologies at the module and cell levels offer superior performance in mitigating cell inconsistency, enabling rapid charging, and improving overall efficiency. Building on these advantages, a variety of battery system topologies with dynamic reconfiguration capabilities have been proposed in recent research. This paper classifies existing RBS topologies according to the number of switches used to control individual cells or modules. It also provides a comprehensive summary of current topologies and presents a detailed introduction to their architectural evolution and specific structural designs.

Additionally, the number of switching devices required for each existing RBS topology is clearly specified in this paper. All analyses related to switching devices use MOSFETs as the reference device. When IGBTs are employed instead, the switch-related results and discussions presented here can serve as a useful reference. The reconfigurable capability of the RBS enables parallel fast-charging modes between batteries during operation. This significantly increases the overall charging rate of the battery system. Furthermore, precise cell-level control allows the RBS to achieve self-balancing through internal topology reconfiguration alone. No external balancing circuits or equipment are required, which greatly improves energy utilization efficiency. The ability to vary inter-battery connection states also permits adjustment of the system's input and output voltage. This provides support for a wide range of voltage levels at both input and output ports. To further characterize the behavior of RBS, this paper adopts an equivalent circuit model to analyze the series-parallel configurations. Particular emphasis is placed on summarizing and evaluating methods for estimating the maximum current during parallel connection. These findings offer practical guidance for the selection of switching devices in engineering applications.

The diversity of RBS topologies offers multiple options for engineering applications. However, different topologies exhibit distinct

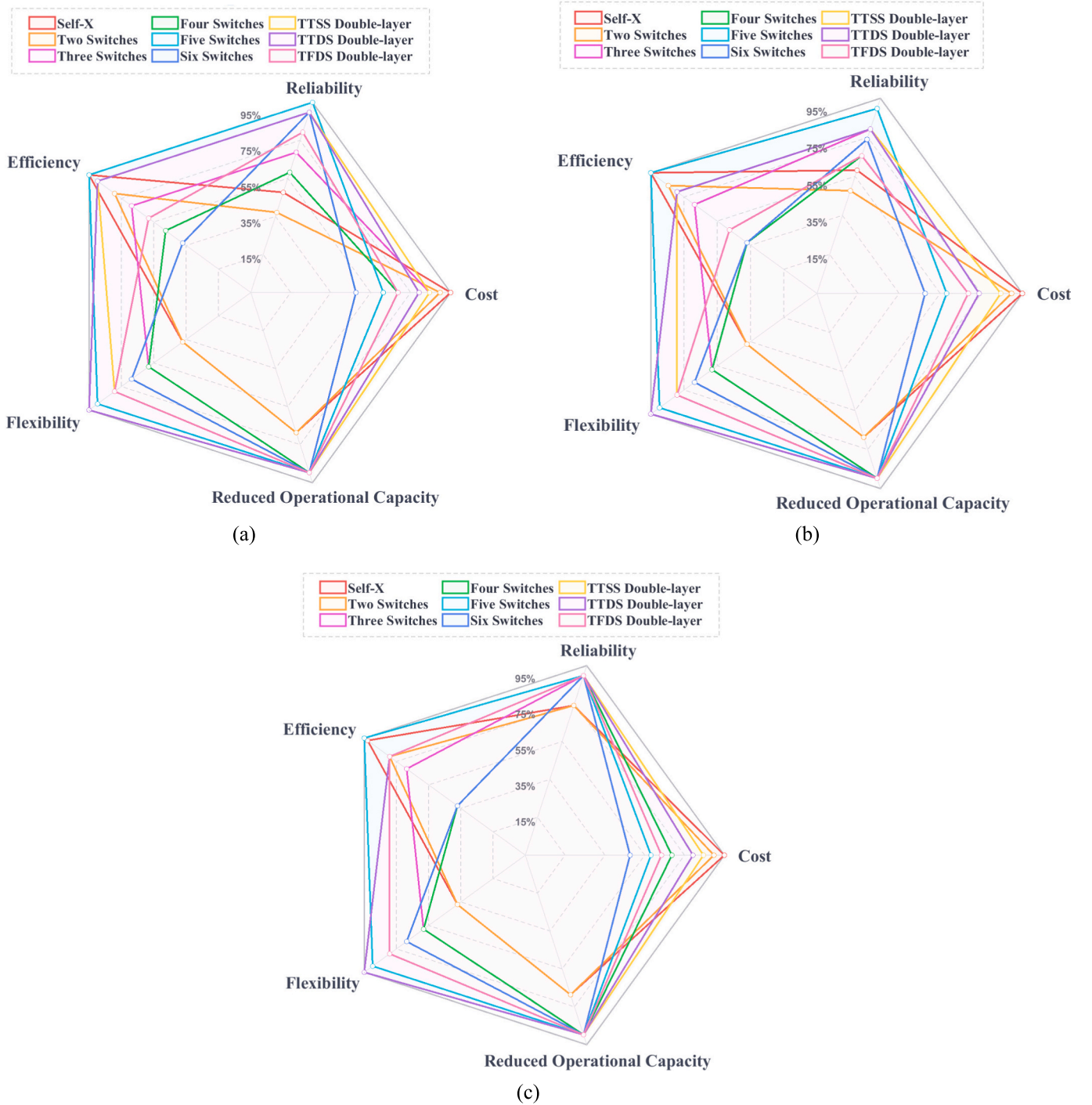


Fig. 39. Comparison chart of the existing topology in terms of cost, efficiency, reliability, flexibility, and reduced operational capacity: (a) Home energy storage scenario. (b) EV energy storage scenario. (c) Drone energy storage scenario.

performance characteristics. The advantages and disadvantages of a given topology can vary significantly depending on the specific application scenario. Simple topologies, such as the Self-X and two-switch configurations, provide clear cost advantages. This results from their use of fewer components and simpler structures. The reduced number of switches also leads to lower switching losses during operation, which contributes to higher efficiency in these designs. Nevertheless, topology efficiency is not determined solely by the number of switches. For example, five-switch topologies can achieve efficiency levels comparable to those of simpler configurations. Similarly, dual-layer topologies often outperform single-layer topologies of the same switch type in

terms of efficiency. Likewise, reliability does not necessarily improve with an increasing number of switches.

It should be noted that simple topologies, despite their limited flexibility, offer low cost and high efficiency. These characteristics make them particularly suitable for small-scale energy storage applications with modest reconfiguration requirements. In medium to large-scale energy storage scenarios, dual-layer topologies are more appropriate. They provide high performance in efficiency, reliability, and flexibility while maintaining significantly lower costs than five-switch topologies. For future research on large-scale energy storage systems, the dual-layer topology can be considered a preferred architectural choice.

CRediT authorship contribution statement

Fanqi Tang: Writing – review & editing, Writing – original draft, Validation, Investigation, Formal analysis, Data curation. **Jianjun Ma:** Writing – review & editing, Supervision, Formal analysis, Conceptualization. **Shuli Wen:** Supervision, Methodology. **Qiang Zhao:** Conceptualization. **Bing Zhang:** Formal analysis. **Yiqi Huang:** Investigation. **Miao Zhu:** Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data was used for the research described in the article.

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